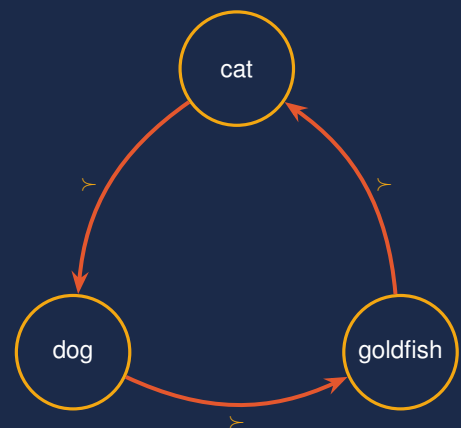


CHAPTER ONE

Choice, Risk & Uncertainty

*or: why your preferences can be pumped for cash, why
Savage failed his own exam, and how many people
you should date before settling down*



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(a money pump; see Section)

Choice, Risk & Uncertainty

Economic theory is often introduced as a collection of models: supply and demand, the firm, the consumer, general equilibrium, each with its own diagram and its own vocabulary. This book treats economic theory instead as a language for asking one recurring question: **what can we conclude once a decision problem has been stated precisely?** A model identifies the alternatives on the table, the agents involved, the information available to them, and the criterion by which outcomes are judged. Theory then tells us which conclusions follow from those ingredients alone, and which further conclusions would require additional value judgments that the model does not supply for free.

This first chapter builds the basic vocabulary of individual choice that the rest of the book will reuse without reintroducing. It moves from preferences over certain alternatives, to lotteries with known probabilities, to genuine uncertainty when probabilities are themselves disputed or unknown, and closes with a short foray into sequential decisions, where information arrives over time and choices cannot be undone. Along the way we will meet several people behaving in ways that look irrational at first glance and turn out, on inspection, to be model-breaking rather than mind-breaking: a philosopher ordering dessert, a Renaissance astronomer choosing a wife, and a man who leaked the Pentagon Papers after publishing an urn problem that changed how economists think about ignorance.

Every result in this chapter comes with a full proof, with one family of exceptions flagged where they occur: the axiomatic foundations of decision under uncertainty in Section 3 (the representation theorems of Savage, of Gilboa and Schmeidler, and of Schmeidler) are stated without proof, as each derivation is a substantial undertaking in its own right. Readers who want the proofs to feel entirely self-contained, without reaching for outside references on order relations, concavity, or elementary real analysis, will find the relevant background collected in Appendix A, which takes only the mean value theorem and the elementary properties of the integral for granted. Everyone else can read straight through.

Caveat · a note on sources

The paradoxes, experiments, and historical results collected in this chapter — Buridan's donkey, the St. Petersburg game, the theorems of von Neumann–Morgenstern and Savage, the Ellsberg and Allais paradoxes, the framing experiments of Tversky and Kahneman, Rabin's calibration theorem, Newcomb's problem, and the secretary problem, among others — are part of the standard, well-documented repertoire of decision theory, and each is attributed to its original source at first mention and in the reference list closing the chapter. The notation, the proofs, the transitions between topics, the numerical illustrations, and the editorial commentary connecting them are the present author's own construction for this book.

1 The rational chooser

1.1 Choice as a formal problem

A decision problem starts with a set of feasible alternatives. Let X denote the set of all conceivable alternatives, and let $B \subseteq X$ denote the **feasible set** — a budget set, or a menu.

The decision maker must select one or more elements of B .

This innocent description already separates three objects that are constantly confused in everyday argument:

- the **alternatives** and their consequences, which is primarily a factual matter;
- the decision maker's **ranking** of those alternatives, which may reflect tastes, ethical commitments, or institutional objectives;
- the **rule** used to translate that ranking into a selection.

The separation is never perfect — the description of an alternative depends on which consequences we decide to measure — but it tells us exactly where assumptions enter an argument, which is often the difference between a productive disagreement and an unproductive one.

Aside · Buridan's donkey

A donkey stands exactly halfway between two identical bales of hay and starves to death for lack of any reason to prefer one to the other. The medieval story is usually told as a paradox of rationality. With the three objects above in hand, it is not one: the donkey's *ranking* is perfectly coherent (it is indifferent between the two bales), and the trouble lies entirely in its *rule*, which fails to select any element from a set of equally good options. Any tie-breaking rule at all, including a coin flip, saves the donkey. Rationality was never the problem; having a complete decision procedure was.

1.2 Preferences

Definition 1.1: Preference relation

A **binary relation** $\succsim$ on X is a set of ordered pairs of alternatives; we read $x \succsim y$ as “ x is at least as good as y ”. The associated strict preference and indifference relations are defined by

$$x \succ y \iff x \succsim y \text{ and not } y \succsim x, \quad x \sim y \iff x \succsim y \text{ and } y \succsim x.$$

Definition 1.2: Rationality

Completeness. For all $x, y \in X$: $x \succsim y$ or $y \succsim x$ (possibly both).

Transitivity. For all $x, y, z \in X$: if $x \succsim y$ and $y \succsim z$, then $x \succsim z$.

A relation that is both complete and transitive is called a **weak order**.

Completeness says that any two alternatives can be compared; it is a demanding requirement whenever alternatives are complex or morally incommensurable.

Aside · A student's dilemma

In *Existentialism is a Humanism*, Jean-Paul Sartre describes a young man in occupied France torn between joining the Free French forces in England and remaining with his mother, for whom he is the only source of comfort. Sartre's point is that no ethical doctrine on offer settles the comparison for him. In our vocabulary: the young man's preference is *incomplete*. Economists have models for incomplete preferences — they simply refuse to rank certain pairs — but the baseline theory in this chapter starts by

assuming incompleteness away, and it is worth remembering, whenever completeness is invoked, that this is a substantive assumption and not a logical necessity.

Transitivity imposes coherence across pairwise comparisons: it forbids the decision maker from preferring x to y , y to z , and z to x all at once. Its best defence is not philosophical but mercenary.

★ Puzzle ★ the money pump

Suppose Kevin prefers a cat to a dog, a dog to a goldfish, and a goldfish to a cat — a preference cycle. Kevin currently owns a cat. A trader offers him the following sequence of trades.

1. Trade the cat for a goldfish, plus €1 from Kevin. Since Kevin strictly prefers the goldfish to the cat, he accepts.
2. Trade the goldfish for a dog, plus €1 from Kevin. He accepts.
3. Trade the dog for a cat, plus €1 from Kevin. He accepts.

Kevin again owns a cat, and is €3 poorer than when he started. Nothing stops the trader from repeating the cycle indefinitely. Cyclic preferences turn their owner into an automated teller machine for whoever discovers the cycle. (The scene is illustrated on the title page.)

Money pumps of this kind will resurface in the study of collective decisions, where a striking fact awaits: majority voting can produce cycles of exactly this kind — Kevin's cat-dog-goldfish loop, played out among candidates or policies — even when every individual voter's own preference is perfectly transitive. The pump can then be run on an entire electorate rather than on one unlucky pet owner.

1.3 Utility: numbers that mean nothing except their order

Definition 1.3: Utility representation

A function $u : X \rightarrow \mathbb{R}$ **represents** the preference relation $\succsim$ if, for all $x, y \in X$,

$$x \succsim y \iff u(x) \geq u(y).$$

The numbers that u assigns need not measure pleasure, satisfaction, or anything psychological at all. If u represents $\succsim$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ is strictly increasing, then $g \circ u$ represents exactly the same preference relation, because g preserves the ordering of real numbers. If a function assigns 10 utils to pizza and 5 to salad, this does *not* mean pizza is liked twice as much as salad: u , u^3 , and e^u all encode precisely the same ranking. Within the ordinal theory developed here, differences and ratios of utility levels carry no meaning whatsoever — only their relative order does.

Aside · utility as intensity of preference

The ordinal reading is a modeling choice, not a dogma. A long tradition, from Bentham and Edgeworth to modern welfare economics, reads utility differences as measuring the *intensity* of preference, and this reading can be given a precise foundation. Suppose the decision maker can compare not only alternatives but also *transitions* between them: going from x to y is a greater improvement than going from z to w . Suitable axioms on

such comparisons (Alt 1936; Suppes and Winet 1955; Krantz, Luce, Suppes, and Tversky 1971) yield a utility function u such that $u(y) - u(x) \geq u(w) - u(z)$ exactly when the first transition is at least as large an improvement as the second, unique up to positive affine transformations: differences, though not levels or ratios, then carry meaning. Whether such a representation is the right one depends on the model and on what it is meant to do. A theory of demand built on observed market choices has no use for it; a theory of collective decisions that weighs one person's gain against another's loss — a utilitarian rule, for instance — cannot do without information of at least this kind. A second, distinct route to cardinal utility appears with the expected utility theorem of Section 2, and we return there to how the two relate.

Theorem 1.4: Representation on finite sets

If X is finite, a preference relation $\succsim$ on X is represented by some function $u : X \rightarrow \mathbb{R}$ if and only if $\succsim$ is complete and transitive.

Proof. ($\Rightarrow$) Suppose u represents $\succsim$. The usual order $\geq$ on $\mathbb{R}$ is complete (any two reals are comparable) and transitive, and both properties transfer back to $\succsim$ through the equivalence $x \succsim y \iff u(x) \geq u(y)$: given x, y , either $u(x) \geq u(y)$ or $u(y) \geq u(x)$ (completeness of $\geq$ on $\mathbb{R}$), hence $x \succsim y$ or $y \succsim x$; and if $u(x) \geq u(y) \geq u(z)$ then $u(x) \geq u(z)$ (transitivity of $\geq$ on $\mathbb{R}$), hence $x \succsim y$ and $y \succsim z$ imply $x \succsim z$.

($\Leftarrow$) Suppose $\succsim$ is complete and transitive on the finite set X . Completeness applied to the pair (x, x) gives $x \succsim x$ or $x \succ x$, so $\succsim$ is **reflexive**, and the indifference relation $\sim$ (defined by $x \sim y$ iff $x \succsim y$ and $y \succsim x$) is an equivalence relation on X : reflexive by what we just showed, symmetric by construction, and transitive because if $x \sim y$ and $y \sim z$ then $x \succsim y \succsim z$ and $z \succsim y \succsim x$, so $x \succsim z$ and $z \succsim x$ by transitivity of $\succsim$.

Let $X/\sim$ denote the set of equivalence classes. Define a relation $\succsim^*$ on $X/\sim$ by $[x] \succsim^* [y]$ iff $x \succsim y$; this is well defined (independent of the representatives x, y chosen) precisely because $\sim$ is the indifference relation of $\succsim$ — replacing x by an indifferent x' , or y by an indifferent y' , preserves the truth of $x \succsim y$ by transitivity. By construction $\succsim^*$ is complete (inherited from $\succsim$) and, crucially, **antisymmetric**: if $[x] \succsim^* [y]$ and $[y] \succsim^* [x]$ then $x \succsim y$ and $y \succsim x$, i.e. $x \sim y$, i.e. $[x] = [y]$. A complete, transitive, antisymmetric relation on a finite set is exactly a linear (total) order, so the n elements of $X/\sim$ can be listed from best to worst as $[x_1] \succsim^* [x_2] \succsim^* \dots \succsim^* [x_n]$ with $[x_i] \neq [x_j]$ for $i \neq j$ — a standard fact proved by induction on n (remove the $\succsim^*$ -worst class, order the rest by the inductive hypothesis, and place the removed class last).

Now define $u : X \rightarrow \mathbb{R}$ by $u(x) = n - i + 1$ whenever $x \in [x_i]$ in this listing, i.e. u assigns the rank of x 's indifference class, counted from the bottom. For $x, y \in X$ with $x \in [x_i], y \in [x_j]$,

$$x \succsim y \iff [x_i] \succsim^* [x_j] \iff i \leq j \iff u(x) \geq u(y),$$

where the middle equivalence holds because the classes are listed from best to worst: $[x_i] \succsim^* [x_j]$ whenever $i \leq j$, while for $i > j$ we have $[x_j] \succsim^* [x_i]$ and $[x_i] \neq [x_j]$, hence not $[x_i] \succsim^* [x_j]$ by antisymmetry. Hence u represents $\succsim$. ■

⚠ Caveat

On infinite sets, completeness and transitivity are no longer sufficient. The standard counterexample is the **lexicographic order** on $\mathbb{R}^2$: declare $(x_1, x_2) \succ (y_1, y_2)$ if $x_1 > y_1$, or $x_1 = y_1$ and $x_2 > y_2$ — compare first coordinates, and break ties with the second. This relation is complete and transitive (checking transitivity is a short case analysis on

whether the first coordinates are equal), yet it admits no utility representation. The idea of the impossibility proof: if u represented this order, then for every real number x_1 we could find a nondegenerate open interval of utility values reserved for alternatives of the form (x_1, x_2) as x_2 ranges over $\mathbb{R}$, and these intervals would have to be pairwise disjoint across distinct values of x_1 . But $\mathbb{R}$ contains only countably many pairwise disjoint nondegenerate intervals (each must contain a distinct rational number, and $\mathbb{Q}$ is countable), while there are uncountably many values of x_1 — a contradiction. A further axiom, usually a continuity requirement on the preference relation, restores representability on infinite sets; this is essentially the content of Debreu's representation theorem, which we do not need in what follows.

1.4 Choice rules and an inconsistent dessert order

Definition 1.5: Choice correspondence, rationalization

A **choice correspondence** assigns to every nonempty feasible set $B \subseteq X$ a nonempty subset $C(B) \subseteq B$, the set of alternatives the decision maker might select from B ; ties are allowed. A **choice function** is a choice correspondence that always selects exactly one alternative. A preference relation $\succsim$ **rationalizes** a choice correspondence C if

$$C(B) = \{x \in B : x \succsim y \text{ for every } y \in B\} \quad \text{for every feasible set } B.$$

This definition separates what we *observe* — a pattern of choices across various menus — from what we *use to explain it* — an underlying, unobserved preference relation. Revealed-preference theory asks precisely when a collection of observed choices can be explained by some coherent ranking. The answer imposes testable restrictions that link choices across different menus; a choice pattern that violates these restrictions cannot come from any stable ranking at all, however the ranking is chosen. The most basic such restriction is the following.

Definition 1.6: Sen's condition α (contraction consistency)

A choice correspondence C satisfies **condition α** if, whenever $x \in C(B)$ and $x \in A \subseteq B$, we also have $x \in C(A)$. In words: an alternative chosen from a menu is still chosen from any smaller menu that contains it — the champion of the world is also the champion of every country it belongs to.

Theorem 1.7: Rationalizability of choice functions

Let X be finite and let C be a choice function defined on every nonempty subset of X . Then C is rationalizable by some preference relation if and only if C satisfies condition α .

Proof. ($\Rightarrow$) Suppose $\succsim$ rationalizes C , so $C(B) = \{x \in B : x \succsim y \forall y \in B\}$ for every B . Let $x \in C(B)$ and $x \in A \subseteq B$. Since $x \in C(B)$, we have $x \succsim y$ for every $y \in B$, and in particular for every $y \in A$ (because $A \subseteq B$). Hence $x \in C(A)$.

($\Leftarrow$) Suppose C satisfies condition α . Since C is a choice *function*, write $c(B)$ for the unique element of $C(B)$, and define a relation $\succsim$ on X by

$$x \succsim y \quad \text{if and only if} \quad x = c(\{x, y\}).$$

(For $x = y$ this gives $x \succsim x$; for $x \neq y$, exactly one of $x \succsim y$ and $y \succsim x$ holds.) We check that $\succsim$ is complete and transitive, and that it rationalizes C .

Completeness. For any $x, y \in X$, $c(\{x, y\})$ equals x or y , so $x \succsim y$ or $y \succsim x$.

Transitivity. Suppose $x \succsim y$ and $y \succsim z$; we show $x \succsim z$. If two of x, y, z coincide, the claim is immediate, so let them be distinct and let $w = c(\{x, y, z\})$. If $w = y$, condition α applied to $\{x, y\} \subseteq \{x, y, z\}$ gives $c(\{x, y\}) = y$, contradicting $x \succsim y$, which says $c(\{x, y\}) = x$. If $w = z$, condition α applied to $\{y, z\} \subseteq \{x, y, z\}$ gives $c(\{y, z\}) = z$, contradicting $y \succsim z$. Hence $w = x$, and condition α applied to $\{x, z\} \subseteq \{x, y, z\}$ gives $c(\{x, z\}) = x$, i.e. $x \succsim z$.

Rationalization. Let $B \subseteq X$ be nonempty, let $x = c(B)$, and let $M(B) = \{z \in B : z \succsim y \text{ for every } y \in B\}$. For every $y \in B$, condition α applied to $\{x, y\} \subseteq B$ gives $c(\{x, y\}) = x$, i.e. $x \succsim y$; hence $x \in M(B)$. Conversely, if $z \in M(B)$ then in particular $z \succsim x$, i.e. $c(\{x, z\}) = z$, whereas we have just shown $c(\{x, z\}) = x$; hence $z = x$. Thus $M(B) = \{x\} = C(B)$ for every B , and $\succsim$ rationalizes C . ■

☞ Caveat · choice correspondences

The theorem is specific to choice *functions*. For choice correspondences, which allow ties, condition α is no longer sufficient. Take $X = \{x, y, z\}$ with $C(\{x, y\}) = \{x, y\}$, $C(\{x, z\}) = \{x\}$, $C(\{y, z\}) = \{y\}$ and $C(\{x, y, z\}) = \{x\}$. Condition α holds (only the two pairs containing x need checking), yet no weak order rationalizes C : the pairwise choices force $x \sim y$ and $y \succ z$, so y would have to be chosen from $\{x, y, z\}$ together with x . What fails is Sen's **condition β** : if $x, y \in C(A)$ and $A \subseteq B$, then $x \in C(B)$ if and only if $y \in C(B)$. Sen (1971) showed that, on a finite set and with C defined on every nonempty subset, a choice correspondence is rationalizable by a weak order if and only if it satisfies both α and β .

★ Puzzle ★ an inconsistent dessert order

The philosopher Sidney Morgenbesser orders dessert. The waitress offers apple pie or blueberry pie. He orders apple. She returns a moment later: "Oh, we also have cherry." Morgenbesser replies: "In that case, I'll have blueberry."

Which axiom did he violate? *Answer:* writing a, b, c for apple, blueberry, cherry, his choices reveal $C(\{a, b, c\}) = \{b\}$ and $b \in \{a, b\} \subseteq \{a, b, c\}$, yet $C(\{a, b\}) = \{a\}$. Condition α fails outright: b was chosen from the larger menu but not from a smaller menu containing it. (The anecdote is probably embellished in the retelling; the logic of the violation is not.)

When an observed choice pattern violates condition α , there are exactly three possible explanations: the feasible set changed in some way not captured by our description of the alternatives, the decision maker's preferences genuinely changed between observations, or the stable-preference model is simply the wrong tool for this situation. Two well-known examples illustrate the first and the third possibility respectively.

Aside · the last apple on the plate

Amartya Sen offers the following example. Offered a fruit basket containing a single apple, a polite guest takes nothing. Offered a basket containing two apples, the same guest takes one. This choice pattern violates condition α on its face — the apple that goes untaken when it is the only option is taken when a second one is available — yet nothing about the guest's behaviour is irrational. The resolution is that the *alternative* "take the last apple in the basket" is not the same consequence as "take one of two apples": the composition of the menu itself carries social information (taking the last apple looks greedy; taking one of two does not), and once this is built into the description of the

alternatives, the apparent violation disappears. Before declaring an axiom violated, it always pays to check exactly what is being held constant across the two choices.

Aside · the decoy subscription

Huber, Payne, and Puto (1982) showed experimentally, across several product categories, that adding an option that is *asymmetrically dominated* — worse than one existing alternative on every attribute, but not worse than the other — raises the share of the alternative that dominates it: the **decoy** or **attraction effect**. Dan Ariely (2008) popularized it with a magazine subscription page offering web access only for \$59, print only for \$125, and print-plus-web also for \$125. In the classroom demonstration he reports, nobody chose print only, 84% chose print-plus-web and the rest web only; with the print-only option removed, the share choosing print-plus-web fell to 32%. These figures come from an informal demonstration rather than a controlled study, and the experimental literature is the better anchor. There, the effect is robust when every attribute is given as a number, but Frederick, Lee, and Baskin (2014) find that it typically vanishes when consumers experience the product or when even one attribute is conveyed perceptually, and they question its practical significance. Where it does occur, the effect resists being explained away by redescribing the alternatives, unlike Sen's apple, and it violates the probabilistic analogue of condition α known as *regularity*: adding an option to a menu should never increase the probability of choosing an option that was already available.

1.5 Several criteria: Pareto efficiency and its limits

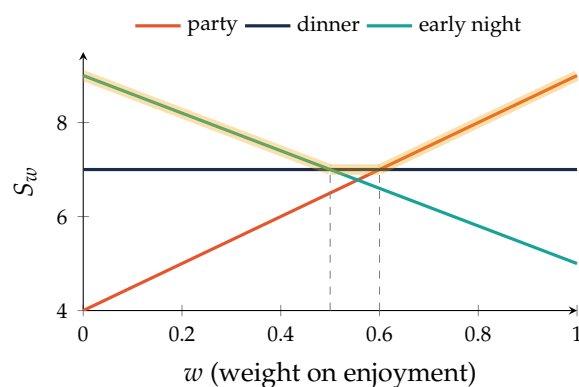
Many decisions involve more than one objective at once. Suppose each alternative x generates a vector of consequences $v(x) = (v_1(x), \dots, v_k(x))$, where a higher number is better on every one of the k criteria.

Definition 1.8: Pareto dominance and efficiency

Alternative x **Pareto dominates** alternative y if $v_j(x) \geq v_j(y)$ for every criterion j , with strict inequality for at least one criterion. An alternative x is **Pareto efficient** in a feasible set B if no feasible alternative Pareto dominates it.

A Pareto-dominated option has an obvious defect: some other feasible option improves on it along at least one dimension without sacrificing anything on any other. But Pareto efficiency is a *screening* principle, and only rarely a complete decision rule on its own, since many alternatives can simultaneously be efficient. Consider an evening with two competing objectives, enjoyment and rest, and three ways to spend it:

Option	Fun v_1	Rest v_2
A: a party	9	4
B: dinner	7	7
C: early night	5	9



None of the three options dominates another, so all three are Pareto efficient — the screening principle eliminates nothing here. A weighted score

$$S_w(x) = w v_1(x) + (1 - w) v_2(x), \quad 0 \leq w \leq 1,$$

makes the trade-off between the objectives explicit and does select a unique winner for almost every value of w :

$$S_w(A) = 4 + 5w, \quad S_w(B) = 7, \quad S_w(C) = 9 - 4w.$$

The early night maximizes the score for $w < 1/2$, dinner with friends for $1/2 < w < 3/5$, and the party for $w > 3/5$, with ties at the two boundary values. Notice that dinner, the intuitive “compromise” option, wins only on the narrow window $w \in (1/2, 3/5)$.

In summary

The calculation above does not discover the “correct” value of w — nothing in the mathematics can, since w encodes a value judgment about how much enjoyment is worth relative to rest, and that judgment is supplied from outside the model. What the calculation shows is precisely *how* the recommended choice depends on the priority assigned to each criterion. This pattern recurs throughout the theory of collective and multi-criterion decisions: a theorem can characterize *every* rule satisfying a certain list of properties, without thereby establishing that those properties are the only defensible ones. Rigor tells us what follows from stated assumptions; it cannot certify the assumptions themselves.

2 Decision under risk

2.1 Lotteries

A decision involves **risk** when its possible consequences *and their associated probabilities* are both known in advance. Let $Z = \{z_1, \dots, z_m\}$ be a finite set of consequences.

Definition 2.1: Lottery, mixture

A **lottery** p on Z assigns a probability $p_i \geq 0$ to each consequence z_i , with $\sum_{i=1}^m p_i = 1$. Write $\Delta(Z)$ for the set of all lotteries on Z ; the degenerate lottery assigning probability one to z_i is identified with the consequence z_i itself. For $p, q \in \Delta(Z)$ and $\alpha \in [0, 1]$, the **mixture** $\alpha p + (1 - \alpha)q$ denotes the lottery obtained by running p with probability α and q with probability $1 - \alpha$.

2.2 Why not simply maximize expected monetary value?

If the consequences are monetary prizes $x_1, \dots, x_m$, the **expected value** of a lottery is $\mathbb{E}_p[X] = \sum_i p_i x_i$. A tempting decision rule is to always choose the lottery with the highest expected value. An eighteenth-century casino in Saint Petersburg would have been delighted to see you walk in.

★ Puzzle ★ how much would you pay to play?

A fair coin is tossed repeatedly until it lands heads for the first time. If the first head appears on toss number k , the payout is 2^k euros — so €2 with probability $1/2$, €4 with probability $1/4$, €8 with probability $1/8$, and so on. What is the largest amount you would be willing to pay for a single play of this game? (Most people, asked this question, answer with a handful of euros.)

The expected value of the game is

$$\mathbb{E}[X] = \sum_{k=1}^{\infty} \frac{1}{2^k} \cdot 2^k = 1 + 1 + 1 + \cdots = +\infty.$$

A strict expected-value maximizer should be willing to sell her house to play this game once. Nobody does. Nicolas Bernoulli posed this puzzle in a 1713 letter; his cousin Daniel Bernoulli supplied the resolution in 1738: people do not value money linearly. Suppose instead of monetary value we evaluate the game using the utility function $u(x) = \log x$. Then

$$\mathbb{E}[\log X] = \sum_{k=1}^{\infty} \frac{1}{2^k} \log(2^k) = \sum_{k=1}^{\infty} \frac{k \log 2}{2^k} = \log 2 \sum_{k=1}^{\infty} \frac{k}{2^k} = \log 2 \cdot 2 = \log 4,$$

using the standard identity $\sum_{k=1}^{\infty} kx^k = x/(1-x)^2$ evaluated at $x = 1/2$, which gives $\sum_k k/2^k = (1/2)/(1/2)^2 = 2$. The game is therefore worth exactly as much, in utility terms, as €4 received for certain. Diminishing marginal utility of money tames an infinite expected value down to a modest certainty equivalent.

Two simplifications are hidden in this computation. Utility is applied to the prize alone, as if the player had no other wealth; and €4 is the certainty equivalent of the prize, not the price the player would pay, since paying to play changes the wealth at which every prize is received. Daniel Bernoulli applied the logarithm to final wealth, in which case the largest acceptable price p for a player with initial wealth w solves $\mathbb{E}[\log(w - p + X)] = \log w$. The answer now depends on w : numerically, about €7.8 for $w = 100$, €11 for $w = 1,000$, and €21 for $w = 1,000,000$ — still modest, and growing only very slowly with wealth.

🔍 Caveat

Logarithmic utility is not a universal cure for this kind of puzzle. The mathematician Karl Menger's (1934) "super-St. Petersburg" variant pays e^{2^k} on toss k instead of 2^k ; then $\mathbb{E}[\log X] = \sum_k 2^{-k} \cdot 2^k = \sum_k 1 = \infty$ once again. In fact, for *every* unbounded utility function u , one can design a lottery with growth rates fast enough to send $\mathbb{E}[u(X)]$ to infinity. This observation is one of the standard arguments in favor of insisting that utility functions be bounded, a technical requirement we will not otherwise dwell on.

2.3 Expected utility: an averaging rule that is derived, not assumed

Given a utility function u over consequences, define the **expected utility** of a lottery p by

$$EU(p) = \sum_{i=1}^m p_i u(z_i).$$

The achievement of John von Neumann and Oskar Morgenstern was to show that this averaging formula is not an arbitrary modeling decision. It *follows*, as a logical consequence, from two restrictions placed directly on preferences over lotteries.

Definition 2.2: Independence

A preference relation $\succsim$ on $\Delta(Z)$ satisfies **independence** if, for all lotteries $p, q, r \in \Delta(Z)$ and every $\alpha \in (0, 1)$,

$$p \succsim q \iff \alpha p + (1 - \alpha)r \succsim \alpha q + (1 - \alpha)r.$$

In words: mixing the same lottery r into two options, with the same probability weight, should never reverse their ranking. If you prefer chocolate ice cream to vanilla, you should still prefer it once the waiter announces a 50% chance that the kitchen serves everyone soup instead, whatever they ordered — the shared possibility of soup does not bear on the comparison between chocolate and vanilla in the other branch. Independence is a powerful assumption, and a controversial one: precisely because its logical consequences are sharp, its empirical violations are highly informative, as we will see later in this chapter.

Definition 2.3: Continuity

A preference relation $\succsim$ on $\Delta(Z)$ satisfies **continuity** if, for any three lotteries $p, q, r \in \Delta(Z)$ ranked $p \succ q \succ r$ (so p is the best of the three, r the worst, and q an intermediate lottery), there exist probabilities $\alpha, \beta \in (0, 1)$ such that

$$\alpha p + (1 - \alpha)r \succ q \succ \beta p + (1 - \beta)r.$$

This is not the most common textbook formulation, which is usually stated as topological continuity of $\succsim$ (the upper and lower contour sets of every lottery are closed in $\Delta(Z)$). Topological continuity implies the mixture-based version, but not conversely in general. In the presence of the other axioms, however, the mixture-based version is all that is needed: Theorem 2.4 below derives from it a representation that is continuous in the topological sense. It is also the version used directly in the proof, which is why we adopt it here. Intuitively, it rules out infinitely abrupt jumps in preference: mixing enough of the worst lottery r into the best lottery p must eventually drag the mixture below q , and mixing enough of p into r must eventually push it above q — no intermediate lottery is infinitely better or worse than every possible mixture of the two extremes. As an intuition pump: would you cross a street to pick up a stray €1 coin, if doing so carried some vanishingly small probability 10^{-12} of being struck and killed by a bus? Almost everyone says yes at some small enough probability, which amounts to accepting continuity with the lottery r standing for the worst outcome (death) mixed in at a tiny weight against the best (life plus a coin) and the intermediate (life, no coin).

Theorem 2.4: Expected utility theorem (von Neumann–Morgenstern)

Let Z be a finite set of consequences. A preference relation $\succsim$ on $\Delta(Z)$ is complete, transitive, continuous, and satisfies independence if and only if there exists a function $u : Z \rightarrow \mathbb{R}$ such that

$$p \succsim q \iff \sum_i p_i u(z_i) \geq \sum_i q_i u(z_i) \quad \text{for all } p, q \in \Delta(Z).$$

Moreover, if u represents $\succsim$ in this sense, so does $au + b$ for any $a > 0$ and any $b \in \mathbb{R}$; and if $\succsim$ is **nontrivial** (some lottery is strictly preferred to another), these are the only functions that represent $\succsim$ in this sense.

Proof. A lottery is a vector of probabilities, so a mixture $\alpha p + (1 - \alpha)q$ is simply a convex

combination of vectors, and mixtures of mixtures obey ordinary algebra; for instance $\beta(\alpha p + (1 - \alpha)q) + (1 - \beta)q = \alpha\beta p + (1 - \alpha\beta)q$. Applying independence in both directions also yields its strict and indifference versions: for $\alpha \in (0, 1)$, $p \succ q$ iff $\alpha p + (1 - \alpha)r \succ \alpha q + (1 - \alpha)r$, and $p \sim q$ iff $\alpha p + (1 - \alpha)r \sim \alpha q + (1 - \alpha)r$.

Necessity. Let $U(p) = \sum_i p_i u(z_i)$. Being induced by a real-valued function, the ranking of lotteries by U is complete and transitive. U is affine, $U(\alpha p + (1 - \alpha)r) = \alpha U(p) + (1 - \alpha)U(r)$, so for $\alpha \in (0, 1)$ comparing $\alpha p + (1 - \alpha)r$ with $\alpha q + (1 - \alpha)r$ amounts to comparing $U(p)$ with $U(q)$: independence holds. If $U(p) > U(q) > U(r)$, the continuous function $\alpha \mapsto \alpha U(p) + (1 - \alpha)U(r)$ runs from $U(r)$ to $U(p)$, so it exceeds $U(q)$ for α close enough to 1 and falls below it for α close enough to 0: continuity holds.

Sufficiency. Assume now that $\succsim$ is complete, transitive, continuous, and satisfies independence.

Step 1: more weight on the better lottery is better. Let $p \succ q$ and $0 \leq \beta < \alpha \leq 1$; we claim $\alpha p + (1 - \alpha)q \succ \beta p + (1 - \beta)q$. Two facts come first. For $\gamma \in (0, 1)$, the strict version of independence with $r = q$ gives $\gamma p + (1 - \gamma)q \succ \gamma q + (1 - \gamma)q = q$; with $r = p$ it gives $p = \gamma p + (1 - \gamma)p \succ \gamma q + (1 - \gamma)p$. Now let $m_\alpha = \alpha p + (1 - \alpha)q$. By the first fact (or trivially if $\alpha = 1$), $m_\alpha \succ q$. A direct computation shows $\beta p + (1 - \beta)q = \gamma q + (1 - \gamma)m_\alpha$ with $\gamma = 1 - \beta/\alpha \in (0, 1]$. If $\gamma = 1$ (that is, $\beta = 0$) this lottery is q and we are done; otherwise the second fact, applied to $m_\alpha \succ q$, gives $m_\alpha \succ \gamma q + (1 - \gamma)m_\alpha$, as claimed.

Step 2: best and worst consequences. Identify each consequence with the degenerate lottery on it. Since Z is finite and $\succsim$ is a weak order, there are $\bar{z}, \underline{z} \in Z$ with $\bar{z} \succsim z \succsim \underline{z}$ for every $z \in Z$. We claim that $\bar{z} \succ p \succ \underline{z}$ for every lottery p , by induction on the number k of consequences to which p gives positive probability. For $k = 1$ this is the choice of $\bar{z}$ and $\underline{z}$. For $k \geq 2$, pick a consequence z with $\pi = p_z \in (0, 1)$ and write $p = \pi z + (1 - \pi)p'$, where p' is p conditioned on the other consequences and therefore involves only $k - 1$ of them. Independence and $\bar{z} \succ z$ give $\pi \bar{z} + (1 - \pi)p' \succ p$; independence and $\bar{z} \succ p'$ (induction hypothesis) give $\bar{z} = (1 - \pi)\bar{z} + \pi \bar{z} \succ (1 - \pi)p' + \pi \bar{z}$. By transitivity, $\bar{z} \succ p$. The argument for $\underline{z}$ is symmetric.

If $\bar{z} \sim \underline{z}$, Step 2 and transitivity make all lotteries indifferent, and any constant function u represents $\succsim$. From now on, assume $\bar{z} \succ \underline{z}$, and write $m(\lambda) = \lambda \bar{z} + (1 - \lambda)\underline{z}$ for $\lambda \in [0, 1]$. By Step 1, $\lambda > \mu$ implies $m(\lambda) \succ m(\mu)$.

Step 3: calibration against best and worst. We claim that for every lottery p there is exactly one number $\lambda(p) \in [0, 1]$ with $p \sim m(\lambda(p))$. Uniqueness follows from Step 1. For existence, take $\lambda = 1$ if $p \sim \bar{z}$ and $\lambda = 0$ if $p \sim \underline{z}$. Otherwise $\bar{z} \succ p \succ \underline{z}$ by Step 2. Let $L = \{\lambda \in [0, 1] : p \succ m(\lambda)\}$, which contains 0, and let $\lambda^* = \sup L$.

- Suppose $m(\lambda^*) \succ p$. Then $\lambda^* > 0$, since $p \succ \underline{z} = m(0)$. As $m(\lambda^*) \succ p \succ \underline{z}$, continuity yields $\alpha \in (0, 1)$ with $\alpha m(\lambda^*) + (1 - \alpha)\underline{z} = m(\alpha \lambda^*) \succ p$. By Step 1 and transitivity, $m(\lambda) \succ p$ for every $\lambda \geq \alpha \lambda^*$, so every element of L is smaller than $\alpha \lambda^*$ and $\sup L \leq \alpha \lambda^* < \lambda^*$: a contradiction.
- Suppose $p \succ m(\lambda^*)$. Then $\lambda^* < 1$, since $\bar{z} \succ p$. As $\bar{z} \succ p \succ m(\lambda^*)$, continuity yields $\beta \in (0, 1)$ with $p \succ \beta \bar{z} + (1 - \beta)m(\lambda^*) = m(\lambda^* + \beta(1 - \lambda^*))$. Thus $\lambda^* + \beta(1 - \lambda^*) \in L$, although it exceeds $\lambda^* = \sup L$: a contradiction.

By completeness, $p \sim m(\lambda^*)$.

Step 4: λ represents $\succsim$ and is affine. By Step 3, transitivity and Step 1, $p \succsim q \iff m(\lambda(p)) \succsim m(\lambda(q)) \iff \lambda(p) \geq \lambda(q)$. Next, for $\alpha \in (0, 1)$, the indifference version of independence,

applied first to $p \sim m(\lambda(p))$ and then to $q \sim m(\lambda(q))$, gives

$$\begin{aligned}\alpha p + (1 - \alpha)q &\sim \alpha m(\lambda(p)) + (1 - \alpha)q \\ &\sim \alpha m(\lambda(p)) + (1 - \alpha)m(\lambda(q)) = m(\alpha\lambda(p) + (1 - \alpha)\lambda(q)),\end{aligned}$$

so, by uniqueness in Step 3, $\lambda(\alpha p + (1 - \alpha)q) = \alpha\lambda(p) + (1 - \alpha)\lambda(q)$.

Step 5: expected utility. Set $u(z) = \lambda(z)$ for $z \in Z$. Writing $p = \pi z + (1 - \pi)p'$ as in Step 2 and applying Step 4, an induction on the number of consequences gives $\lambda(p) = \sum_i p_i u(z_i)$ for every lottery p . Combined with Step 4, $p \succsim q \iff \sum_i p_i u(z_i) \geq \sum_i q_i u(z_i)$. Note that $u(\bar{z}) = 1$ and $u(\underline{z}) = 0$.

Step 6: uniqueness. That $au + b$ represents $\succsim$ when $a > 0$ is immediate, since $\sum_i p_i (au(z_i) + b) = a \sum_i p_i u(z_i) + b$. Conversely, let $\succsim$ be nontrivial and let v represent it in expected-utility form, with $V(p) = \sum_i p_i v(z_i)$. Nontriviality rules out $\bar{z} \sim \underline{z}$, so $\bar{z} \succ \underline{z}$ and $v(\bar{z}) > v(\underline{z})$. For every $z \in Z$, Step 3 gives $z \sim m(u(z))$, and indifferent lotteries have equal values under V ; hence

$$v(z) = V(m(u(z))) = u(z)v(\bar{z}) + (1 - u(z))v(\underline{z}) = au(z) + b,$$

with $a = v(\bar{z}) - v(\underline{z}) > 0$ and $b = v(\underline{z})$. ■

This uniqueness result marks a genuine change in kind from ordinary ordinal utility: within expected-utility theory, differences in utility now carry meaning, exactly as differences in temperature do. The Celsius and Fahrenheit scales are related by a positive affine transformation, and the statement “today is warmer than yesterday by more than yesterday was warmer than the day before” is meaningful in either scale (both scales agree on whether it is true); the statement “today is twice as hot as yesterday” is meaningless in both. Bernoulli utility functions behave the same way: utility *differences* — and hence comparisons of how much better one gap is than another — are preserved across the admissible rescalings, while utility *levels* and *ratios* are not.

What do these cardinal differences measure? Strictly speaking, the theorem says only that they encode how the decision maker trades off probabilities of better and worse consequences — her attitude toward risk. Whether they also measure the intensity of her preferences between sure consequences, in the sense of the aside of Section 1.3, is a separate question, and the answer depends on the model. Some authors have insisted on keeping the two notions apart (Ellsberg 1954; Baumol 1958). Others identify them, most influentially Harsanyi (1955), whose utilitarian aggregation of individual von Neumann–Morgenstern utilities treats them as measures of individual welfare. When the decision maker’s attitude toward risk is taken to reflect nothing but how much more she values one consequence than another, the identification is entirely appropriate, and the same function measures both; when risk attitudes are thought to have a source of their own — a taste for or against gambling as such — the two scales can come apart.

2.4 Attitudes toward risk

For the remainder of this section, let consequences be levels of final wealth, and let u be increasing (more money is always weakly better).

Definition 2.5: Risk aversion

Let $\succsim$ be a preference relation over lotteries whose consequences are wealth levels, and for a lottery X write $\delta_{E[X]}$ for the degenerate lottery paying its expected value for certain.

The decision maker is **risk averse** if $\delta_{\mathbb{E}[X]} \succ X$ for every lottery X : no lottery is strictly preferred to receiving its own expected value for certain. She is **risk neutral** if $\delta_{\mathbb{E}[X]} \sim X$ for every X , and **risk loving** if $X \succ \delta_{\mathbb{E}[X]}$ for every X .

The definition refers to preferences only, not to any utility function, and applies whether or not preferences have the expected-utility form. Under expected utility, it becomes a property of the Bernoulli utility function.

Theorem 2.6: Risk aversion equals concavity

Under expected utility, a decision maker with Bernoulli utility u is risk averse if and only if u is concave; risk neutral if and only if u is affine; risk loving if and only if u is convex.

Proof. Under expected utility, $\delta_{\mathbb{E}[X]} \succ X$ holds if and only if $u(\mathbb{E}[X]) \geq \mathbb{E}[u(X)]$, so risk aversion is the requirement that $u(\mathbb{E}[X]) \geq \mathbb{E}[u(X)]$ for every lottery X . We prove the risk-averse case; the other two are identical with the inequalities reversed or replaced by equality.

($\Leftarrow$) Suppose u is concave. By definition, concavity means that for any wealth levels $x_1, \dots, x_m$ and any weights $p_1, \dots, p_m \geq 0$ summing to one,

$$u\left(\sum_i p_i x_i\right) \geq \sum_i p_i u(x_i).$$

(This is exactly Jensen's inequality for concave functions; see Appendix A.3 for the definition of concavity and a proof of this inequality by induction on m , starting from the two-point case that defines concavity directly.) The left-hand side is $u(\mathbb{E}[X])$ and the right-hand side is $\mathbb{E}[u(X)]$ for the lottery X taking value x_i with probability p_i . Hence $u(\mathbb{E}[X]) \geq \mathbb{E}[u(X)]$ for every lottery X , which is exactly risk aversion.

($\Rightarrow$) Suppose the decision maker is risk averse, so $u(\mathbb{E}[X]) \geq \mathbb{E}[u(X)]$ for every lottery X . Apply this to the specific two-point lottery that yields wealth x with probability λ and wealth y with probability $1 - \lambda$, for arbitrary x, y and $\lambda \in [0, 1]$. Its expected value is $\lambda x + (1 - \lambda)y$, so risk aversion gives

$$u(\lambda x + (1 - \lambda)y) \geq \lambda u(x) + (1 - \lambda)u(y),$$

which is precisely the definition of concavity of u (see Appendix A.3). Since x, y, λ were arbitrary, u is concave. ■

A worked numerical example

Suppose your final wealth will be either €50 or €150, each with probability 1/2; the expected wealth is €100. Take the Bernoulli utility function $u(x) = \sqrt{x}$, which is concave (its second derivative is $-\frac{1}{4}x^{-3/2} < 0$ for $x > 0$; see Appendix A.3 on how the sign of the second derivative determines concavity). Then

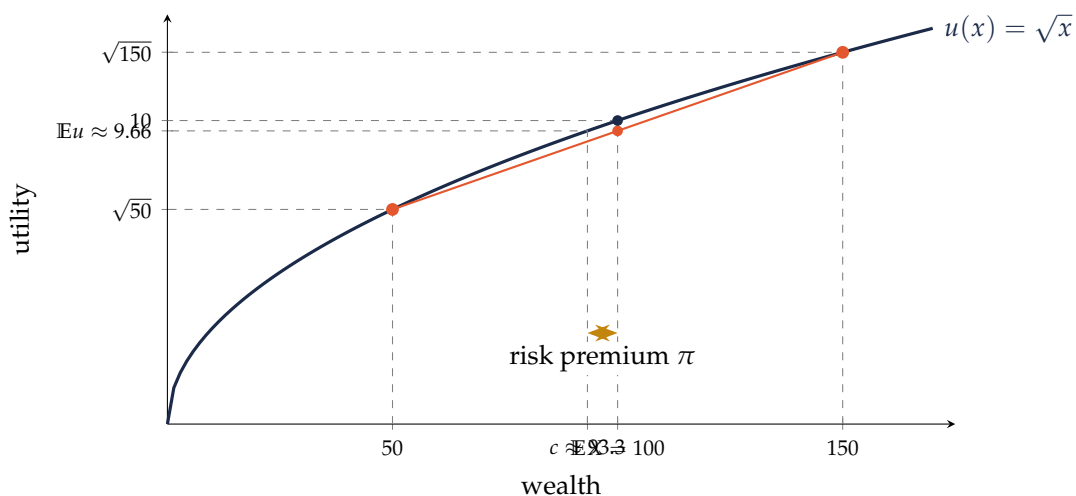
$$\frac{1}{2}\sqrt{50} + \frac{1}{2}\sqrt{150} \approx 9.66 < 10 = \sqrt{100}.$$

The sure wealth of €100 is preferred to the lottery, even though both have the same expected value. Intuitively, the downside branch of the lottery costs more in utility terms than the upside branch gains, precisely because u bends downward as wealth grows.

Definition 2.7: Certainty equivalent and risk premium

The **certainty equivalent** of a lottery X is the sure wealth level $c(X)$ satisfying $u(c(X)) = \mathbb{E}[u(X)]$. The **risk premium** is $\pi(X) = \mathbb{E}[X] - c(X)$: the amount of expected wealth the decision maker would sacrifice to replace the lottery by its certainty equivalent. For a risk-averse decision maker, $\pi(X) \geq 0$ always, since $u(c(X)) = \mathbb{E}[u(X)] \leq u(\mathbb{E}[X])$ and u is increasing.

In the numerical example, $\mathbb{E}[u(X)] \approx 9.66$, so $c(X) \approx 9.66^2 \approx 93.3$, and the risk premium is $\pi \approx 100 - 93.3 = 6.7$ euros — the most this particular decision maker would pay to be rid of the risk entirely.

**Aside · a game show as a laboratory**

In the television game show *Deal or No Deal*, a contestant holds a sealed box containing anywhere from one cent to a small fortune, while a “banker” repeatedly offers a sure sum of money to buy the box back. The banker’s offer functions as a proposed certainty equivalent, and a contestant’s decision to accept it reveals that her risk premium exceeds the gap between the box’s expected value and the offer. Post, van den Assem, Baltussen, and Thaler (2008) used broadcast episodes of the show as a large-stakes laboratory for studying risk attitudes. They found that willingness to gamble depended strongly on recent history within the same game: contestants who had just seen large sums eliminated from contention — a run of bad luck — became *more* willing to take further gambles, not less. We will return to why this pattern is awkward for the theory in Section 4.3.

Measuring the curvature of utility

For a twice-differentiable utility function u with $u' > 0$, define the **Arrow–Pratt coefficients** of absolute and relative risk aversion:

$$A(w) = -\frac{u''(w)}{u'(w)}, \quad R(w) = -w \frac{u''(w)}{u'(w)}.$$

Both coefficients are invariant under positive affine transformations of u — as they must be, since Theorem 2.4 tells us such transformations carry no behavioural content. (Direct check: replacing u by $au + b$ replaces u'' by au'' and u' by au' , and the factor of a cancels in the

ratio.) The coefficient A is the natural measure for risks expressed in absolute monetary units; R is natural for risks expressed as a proportion of current wealth. Two standard families: $u(w) = -e^{-aw}$ has constant absolute risk aversion $A \equiv a$ (CARA); $u(w) = w^{1-\rho}/(1-\rho)$ has constant relative risk aversion $R \equiv \rho$ (CRRA), with $\sqrt{w}$ corresponding to $\rho = 1/2$ and $\log w$ to the limiting case $\rho = 1$.

Theorem 2.8: Pratt's small-risk approximation

Let u be twice continuously differentiable near w , with $u'(w) > 0$, and let ε be a zero-mean risk taking finitely many values, with variance σ^2 . For $t > 0$, let $\pi(t)$ be the risk premium of the scaled risk $t\varepsilon$ at wealth w , defined by $u(w - \pi(t)) = \mathbb{E}[u(w + t\varepsilon)]$. Then

$$\lim_{t \rightarrow 0} \frac{\pi(t)}{t^2} = \frac{1}{2} \sigma^2 A(w), \quad \text{that is,} \quad \pi(t) = \frac{1}{2} \text{Var}(t\varepsilon) A(w) + o(t^2).$$

Proof. If $\sigma^2 = 0$ then $\pi \equiv 0$, so assume $\sigma^2 > 0$; write $e_1, \dots, e_m$ for the values of ε and $p_1, \dots, p_m$ for their probabilities. Near w , $u' > 0$, so u is continuous and strictly increasing there. Since $\mathbb{E}[\varepsilon] = 0$, the smallest value $e_{\min}$ is negative and the largest $e_{\max}$ positive, and for small t the number $\mathbb{E}[u(w + t\varepsilon)]$ lies between $u(w + te_{\min})$ and $u(w + te_{\max})$. By the intermediate value theorem there is a unique $\pi(t)$ with $u(w - \pi(t)) = \mathbb{E}[u(w + t\varepsilon)]$, and $|\pi(t)| \leq t \max_i |e_i|$; in particular $\pi(t) \rightarrow 0$.

Taylor's theorem (Appendix A.4), applied to each of the finitely many terms, gives

$$\mathbb{E}[u(w + t\varepsilon)] = \sum_i p_i \left(u(w) + te_i u'(w) + \frac{1}{2} t^2 e_i^2 u''(w) + o(t^2) \right) = u(w) + \frac{1}{2} t^2 \sigma^2 u''(w) + o(t^2),$$

using $\sum_i p_i e_i = 0$ and $\sum_i p_i e_i^2 = \sigma^2$. On the other side, differentiability of u at w gives $u(w - \pi(t)) = u(w) - \pi(t)(u'(w) + \eta(t))$ with $\eta(t) \rightarrow 0$, because $\pi(t) \rightarrow 0$. Equating the two expressions and dividing by t^2 ,

$$\frac{\pi(t)}{t^2} (u'(w) + \eta(t)) = -\frac{1}{2} \sigma^2 u''(w) + o(1),$$

and letting $t \rightarrow 0$ gives $\pi(t)/t^2 \rightarrow -\frac{1}{2} \sigma^2 u''(w)/u'(w) = \frac{1}{2} \sigma^2 A(w)$. ■

As a check against the exact answer above, use the approximation $\pi \approx \frac{1}{2} \text{Var} A(w)$ directly: $\text{Var} = 50^2 = 2500$ (the risk is ± 50 around the mean of 100, each with probability $1/2$), $A(100) = 1/(2 \cdot 100) = 0.005$ for $u = \sqrt{\cdot}$, giving $\pi \approx 6.25$, reasonably close to the exact value of 6.7 computed earlier even though this particular risk — plus or minus half of expected wealth — is far from “small.”

2.5 Insurance

Let initial wealth be w and let $L \geq 0$ be a random loss. Without insurance, final wealth is the random variable $w - L$. Full insurance at the **actuarially fair** premium $\mathbb{E}[L]$ replaces this with the certain final wealth $w - \mathbb{E}[L]$. If u is concave, Jensen's inequality (Appendix A.3) gives directly

$$u(w - \mathbb{E}[L]) \geq \mathbb{E}[u(w - L)].$$

A risk-averse expected-utility maximizer therefore weakly prefers full insurance at a fair price to bearing the loss uninsured. Real-world insurance contracts include administrative loading, deductibles, exclusions, asymmetric information between insurer and insured, and market

power on one or both sides; the fair-premium benchmark isolates, in the cleanest possible setting, the pure value of eliminating risk, against which all of these real-world frictions can be measured.

Aside · a puzzle about the same people

The same individuals routinely buy home insurance, revealing risk aversion, and also buy lottery tickets, revealing risk-loving behaviour — both at prices that are actuarially unfair to the buyer. Milton Friedman and Leonard Savage (1948) proposed resolving this by allowing a utility function that is concave over some ranges of wealth and convex over others — concave, then convex, then concave again, as wealth increases. Markowitz (1952) soon noted an awkward implication: such a function predicts that individuals in the convex region — often modeled as the middle of the wealth distribution — should be eager to accept enormous gambles, a prediction that sits uneasily with how insurance and gambling actually coexist among the same people. The puzzle has occupied behavioural economists ever since, and it is worth remembering, the next time a phone retailer offers an extended warranty, that such warranties are almost always a bad deal in expected-value terms — which is exactly why they are profitable to sell.

3 Uncertainty and ambiguity

Risk, as defined above, presupposes that probabilities are known. Most decisions that matter in practice do not come with probabilities attached: will a particular start-up survive its first five years, is a given climate model well calibrated, will a business partner honor an informal agreement? These are questions about **uncertainty** rather than risk — a distinction usually traced to Frank Knight (1921) — and it turns out to run deeper than it first appears.

Definition 3.1: Acts and subjective expected utility

Let S be a set of states of the world and Z a set of consequences. An **act** is a function $f : S \rightarrow Z$ specifying which consequence obtains in each state. Leonard Savage (1954) showed that a preference relation over acts satisfying a suitable list of axioms — analogues, for acts, of completeness, transitivity, and a sure-thing principle we meet below — can be represented, jointly, by a probability distribution P over S and a utility function u over Z , in the sense that acts are ranked by the **subjective expected utility**

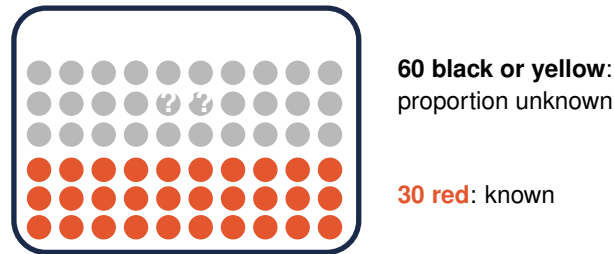
$$V(f) = \sum_{s \in S} P(s) u(f(s)).$$

Crucially, neither P nor u is assumed at the outset: both are derived together, as a pair, from the axioms on preferences over acts. P is called the decision maker's **subjective** probability distribution over states.

The formula looks identical to expected utility under objective risk, but its interpretation is different: the probabilities $P(s)$ are no longer given by the physics of a coin or the rules of a lottery — they represent the decision maker's personal *beliefs*, and they are recovered from behaviour rather than posited in advance. This is a substantially harder representation theorem than von Neumann–Morgenstern's, precisely because it must pin down beliefs and tastes simultaneously from a single set of observed choices over acts, with no independently given notion of probability to lean on; we state it here without proof. A natural question follows immediately: is a single, additive subjective probability distribution always rich

enough to describe how a reasonable person responds to unknown odds?

3.1 An urn with unknown proportions



Consider an urn holding 90 balls. Exactly 30 are red. The remaining 60 are black or yellow, but in a proportion that is entirely unspecified — it could be all black, all yellow, or anything in between. One ball will be drawn at random. Compare the following two pairs of bets, each pair offering the same prize for a win.

	Bet	Wins if the ball drawn is		Common choice
Pair 1	f_1 vs g_1	red	vs black	f_1 (red)
Pair 2	f_2 vs g_2	red or yellow	vs black or yellow	g_2 (black or yellow)

★ Puzzle ★ a contradiction hiding in plain sight

In the experiments associated with this paradox, most subjects, asked to choose in both pairs, choose red in Pair 1 and black-or-yellow in Pair 2. Try to find a single probability distribution over {red, black, yellow} consistent with both choices before reading on.

No such distribution exists — that is, no single additive subjective probability distribution can represent both choices at once, whatever the decision maker's beliefs about the urn happen to be. Write $P(R)$, $P(B)$, $P(Y)$ for the (unknown, subjective) probabilities of red, black, and yellow. Since the prize is identical across all four bets, the decision maker's utility function plays no role in the comparison, and preferring f_1 to g_1 under subjective expected utility requires $P(R) > P(B)$. Preferring g_2 to f_2 requires $P(B) + P(Y) > P(R) + P(Y)$, which simplifies to $P(B) > P(R)$ — the exact opposite inequality. No single additive probability distribution can satisfy both simultaneously.

The structural point is worth isolating: Pair 2 is obtained from Pair 1 simply by adding the same consequence (winning if yellow is drawn) to both bets. Savage's own axioms include a principle, the **sure-thing principle**, stating precisely that a common consequence added identically to both options in a comparison should never change the ranking — an axiom playing exactly the role that independence plays for objective lotteries. In the experiments associated with this paradox, the majority pattern described above (red in Pair 1, black-or-yellow in Pair 2) recurs reliably, showing that this majority routinely violates the sure-thing principle in exactly this setting.

The standard interpretation is **ambiguity aversion**: a preference for bets carrying known odds over otherwise identical bets carrying poorly specified odds. The example does not show that reasoning under genuine uncertainty is irrational — both choices individually look entirely sensible — it shows that a single additive subjective probability distribution is not a rich enough object to capture every coherent attitude toward uncertainty.

Aside · the man behind the urn

Daniel Ellsberg published this paradox in 1961 while working as a defense analyst at the RAND Corporation. A decade later, he leaked the Pentagon Papers — the classified internal history of United States decision-making in Vietnam — to the press, and was charged under the Espionage Act (the charges were eventually dismissed on grounds of governmental misconduct). Few economists have become famous twice over for two such different reasons. His doctoral dissertation, written on the very same topic as the paradox that bears his name, was not published as a book until 2001, forty years after the original article.

3.2 Maxmin expected utility

One influential response to the Ellsberg pattern replaces a single subjective probability with a whole *set* of probabilities the decision maker considers plausible, and evaluates each act by its worst-case expected utility across that set.

Definition 3.2: Maxmin expected utility

Let $\mathcal{P}$ be a set of probability distributions over the state space S . An act f is evaluated by

$$V(f) = \min_{P \in \mathcal{P}} \mathbb{E}_P[u(f)].$$

This decision rule was axiomatized by Itzhak Gilboa and David Schmeidler (1989): consider several probabilistic models of the world at once, and judge each available act by how it performs under the model that is least favourable to it.

Applying maxmin to the Ellsberg urn. Normalize $u(\text{win}) = 1$ and $u(\text{lose}) = 0$, and take the set of plausible distributions to be

$$\mathcal{P} = \left\{ P : P(R) = \frac{1}{3}, P(B) = b, P(Y) = \frac{2}{3} - b, b \in [0, \frac{2}{3}] \right\}$$

— reflecting that the 30-out-of-90 proportion of red is known exactly, while the split between black and yellow among the remaining 60 balls is left entirely open. Then

$$V(f_1) = \frac{1}{3}, \quad V(g_1) = \min_b b = 0, \quad V(f_2) = \min_b (1 - b) = \frac{1}{3}, \quad V(g_2) = \frac{2}{3}.$$

Hence $f_1 \succ g_1$ and $g_2 \succ f_2$: exactly the pattern most people display, produced here by a perfectly coherent decision rule rather than by any inconsistency. Maxmin expected utility captures a cautious, worst-case-oriented attitude toward unknown odds, though it can be criticized as excessively pessimistic in some applications — a decision maker who always expects nature to hand her the single worst distribution in $\mathcal{P}$ is adopting a rather paranoid view of the world, and richer models (allowing, for instance, a weighted average of the worst and the best case) have been developed to soften this.

It is worth being explicit about what this decision rule is and is not. Maxmin expected utility is one normative response to the Ellsberg pattern among several competing ones in the literature, not a uniquely forced conclusion from the failure of a single subjective prior. It encodes a specific, strongly cautious attitude toward ambiguity — replacing one prior with the pessimistic evaluation of a whole set of priors is itself a normative choice, not a logical necessity — and Gilboa and Schmeidler’s own axiomatization — set in the framework of Anscombe and Aumann (1963), where acts yield lotteries — weakens independence to *certainly independence* (mixing two acts with the same constant act never reverses their ranking) and adds an axiom

of *uncertainty aversion* (a mixture of two indifferent acts is weakly preferred to each of them, because mixing hedges); these axioms are no less open to challenge than Savage's original single-prior axioms were. Nor is the rule guaranteed to match every observed ambiguity-averse choice pattern beyond the specific Ellsberg comparison it was built to rationalize. The lesson drawn at the end of Section 1.5 applies here as much as anywhere: choosing $\mathcal{P}$ and the worst-case criterion is itself a modeling decision with content, not a hypothesis-free repair of expected utility.

3.3 Non-additive beliefs: the Choquet integral

Maxmin expected utility gives up the single prior. A second response to Ellsberg, due to David Schmeidler (1989), keeps a single object describing the decision maker's beliefs but gives up *additivity*: the weight attached to the union of two disjoint events need not be the sum of their weights.

Definition 3.3: Capacity

Let S be a finite set of states. A **capacity** on S assigns to every event $E \subseteq S$ a number $v(E) \in [0, 1]$, with $v(\emptyset) = 0$, $v(S) = 1$, and $v(E) \leq v(F)$ whenever $E \subseteq F$. The capacity is **convex** if

$$v(E \cup F) + v(E \cap F) \geq v(E) + v(F) \quad \text{for all events } E, F \subseteq S.$$

A probability distribution is a capacity for which this inequality always holds with equality. For the Ellsberg urn, a natural capacity records, for each event, the probability that is guaranteed whatever the composition of the urn:

$$v(R) = \frac{1}{3}, \quad v(B) = v(Y) = 0, \quad v(R \cup B) = v(R \cup Y) = \frac{1}{3}, \quad v(B \cup Y) = \frac{2}{3},$$

with $v(\emptyset) = 0$ and $v(S) = 1$. Black and yellow separately receive no weight at all, since either colour could be absent from the urn, while together they are known to make up two thirds of it. Additivity fails, as it should: $v(B) + v(R \cup Y) = \frac{1}{3} < 1$, and the missing two thirds measure, in a sense, how little the decision maker knows. Formally, $v(E) = \min_{P \in \mathcal{P}} P(E)$ for every event E , where $\mathcal{P}$ is the set of distributions of Section 3.2; and v is convex (Exercise E8).

How should an act be evaluated against a capacity? The trick is to rewrite expected utility so that it never adds up the probabilities of disjoint events. Let $g : S \rightarrow [0, \infty)$ give the utility obtained in each state, and order the states $s_1, \dots, s_n$ so that $g(s_1) \geq g(s_2) \geq \dots \geq g(s_n)$; set $g(s_{n+1}) = 0$ and $A_i = \{s_1, \dots, s_i\}$, the event on which utility is at least $g(s_i)$. For a probability distribution P , regrouping the terms of the sum gives

$$\sum_{i=1}^n P(s_i) g(s_i) = \sum_{i=1}^n (g(s_i) - g(s_{i+1})) P(A_i).$$

(Expand $P(A_i) = \sum_{j \leq i} P(s_j)$ and exchange the order of summation: the coefficient of $P(s_j)$ is $\sum_{i \geq j} (g(s_i) - g(s_{i+1})) = g(s_j)$.) The right-hand side reads: the act yields at least $g(s_n)$ for sure, an increment $g(s_{n-1}) - g(s_n)$ on the event A_{n-1} , and so on, each increment weighted by the probability of the event on which it is obtained. Unlike the left-hand side, it makes sense for any capacity.

Definition 3.4: Choquet integral, Choquet expected utility

Let v be a capacity on S and $g : S \rightarrow [0, \infty)$, with the states ordered as above. The **Choquet integral** of g with respect to v is

$$\int g dv = \sum_{i=1}^n (g(s_i) - g(s_{i+1})) v(A_i), \quad g(s_{n+1}) = 0.$$

With utility normalized so that $u \geq 0$, the **Choquet expected utility** of an act f is $V(f) = \int u \circ f dv$.

The value does not depend on how states with equal utility are ordered: swapping two such adjacent states changes only a set A_i whose coefficient $g(s_i) - g(s_{i+1})$ is zero. When v is a probability, the Choquet integral is the ordinary expectation. The integral goes back to Choquet (1954). Schmeidler (1989) characterized Choquet expected utility, in the framework of Anscombe and Aumann (1963), by restricting independence to *comonotonic* acts — acts that rank the states in the same order, so that mixing them never hedges one against the other — and Gilboa (1987) obtained a characterization within Savage's framework. We state these results without proof.

Applying Choquet expected utility to the Ellsberg urn. For a bet paying utility 1 on an event E and 0 elsewhere, the definition gives simply $V = v(E)$. Hence

$$\begin{aligned} V(f_1) &= v(R) = \frac{1}{3} > 0 = v(B) = V(g_1), \\ V(f_2) &= v(R \cup Y) = \frac{1}{3} < \frac{2}{3} = v(B \cup Y) = V(g_2), \end{aligned}$$

exactly the Ellsberg pattern. The computation also shows *where* the sure-thing principle gives way. Passing from Pair 1 to Pair 2 adds the same consequence — a win if yellow is drawn — to both bets. Under an additive prior, this adds the same amount $P(Y)$ to both values. Under the capacity, it adds $v(R \cup Y) - v(R) = 0$ to the bet on red but $v(B \cup Y) - v(B) = \frac{2}{3}$ to the bet on black: a win on yellow is worth nothing in combination with red, but it complements black, because black and yellow together form a known two-thirds chance. Non-additivity is the formal expression of this complementarity.

The two answers to Ellsberg — a set of priors and a non-additive capacity — are closely related.

Theorem 3.5: Convex capacities and maxmin expected utility

Let v be a convex capacity on a finite set S , and let its **core** be the set of probability distributions that give every event at least its capacity,

$$\text{core}(v) = \{P : P(E) \geq v(E) \text{ for every } E \subseteq S\}.$$

Then, for every $g : S \rightarrow [0, \infty)$,

$$\int g dv = \min_{P \in \text{core}(v)} \mathbb{E}_P[g].$$

In particular, Choquet expected utility with a convex capacity is maxmin expected utility with the set of priors $\mathcal{P} = \text{core}(v)$.

Proof. Order the states so that $g(s_1) \geq \dots \geq g(s_n)$, and let $A_0 = \emptyset$ and $A_i = \{s_1, \dots, s_i\}$.

No distribution in the core does worse than the Choquet integral. If $P \in \text{core}(v)$, the regrouped

form of $\mathbb{E}_P[g]$ and the nonnegativity of the coefficients $g(s_i) - g(s_{i+1})$ give

$$\mathbb{E}_P[g] = \sum_{i=1}^n (g(s_i) - g(s_{i+1}))P(A_i) \geq \sum_{i=1}^n (g(s_i) - g(s_{i+1}))v(A_i) = \int g dv.$$

Some distribution in the core attains it. Define P^* by $P^*(s_i) = v(A_i) - v(A_{i-1})$. These numbers are nonnegative, by monotonicity of v , and sum to $v(S) = 1$; so P^* is a probability distribution, with $P^*(A_i) = v(A_i)$ for every i , and therefore $\mathbb{E}_{P^*}[g] = \int g dv$. It remains to show that $P^* \in \text{core}(v)$, and this is where convexity enters. Fix an event E . For each i with $s_i \in E$, the events A_{i-1} and $E \cap A_i$ have union A_i and intersection $E \cap A_{i-1}$, so convexity gives $v(A_i) + v(E \cap A_{i-1}) \geq v(A_{i-1}) + v(E \cap A_i)$, that is,

$$P^*(s_i) \geq v(E \cap A_i) - v(E \cap A_{i-1}).$$

For $s_i \notin E$, the right-hand side is zero, since then $E \cap A_i = E \cap A_{i-1}$. Summing,

$$P^*(E) = \sum_{i: s_i \in E} P^*(s_i) \geq \sum_{i=1}^n (v(E \cap A_i) - v(E \cap A_{i-1})) = v(E \cap S) - v(\emptyset) = v(E).$$

Hence $P^* \in \text{core}(v)$, and the minimum is attained at P^* . ■

For the Ellsberg capacity, the core is exactly the set $\mathcal{P}$ of Section 3.2: the constraints $P(B \cup Y) \geq \frac{2}{3}$ and $P(R) \geq \frac{1}{3}$ force $P(R) = \frac{1}{3}$, after which the others hold automatically. The Choquet and maxmin analyses of the urn are therefore one and the same. In general, however, neither model contains the other: Choquet expected utility with a capacity that is not convex is not of the maxmin form, and on larger state spaces there are sets of priors that are not the core of any convex capacity.

Aside · a capacity is a cooperative game

Readers who know cooperative game theory will have recognized the objects in this proof. Read states as players and events as coalitions: a capacity is then a transferable-utility game normalized so that the grand coalition is worth 1; the core of the capacity is the core of the game, the set of divisions of the grand coalition's worth that no coalition can improve upon on its own; and convexity is the convexity of games introduced by Lloyd Shapley (1971). The distribution P^* of the proof is a *marginal contribution vector*: each state receives what it adds to the coalition of the states ranked above it. The second half of the proof is Shapley's theorem that the marginal vectors of a convex game lie in its core; Shapley showed, moreover, that the core of a convex game is exactly the convex hull of its marginal vectors. The same mathematics thus describes how a group can share a surplus without any coalition objecting, and how a decision maker can hold beliefs that are not additive.

4 Putting the model to the test

A normative model of decision-making can remain useful even when observed behaviour violates some of its axioms — indeed, that is often exactly when the model teaches us the most. Every violation forces us to separate three distinct questions.

1. Does the axiom in question actually describe how people behave?

2. Does the axiom express a compelling standard of what *coherent* choice should look like, whether or not people live up to it?
3. If behaviour violates the axiom, should we revise the model, the specific decision at hand, or our assessment of the behaviour itself?

The four examples in this section were each designed, with more or less mathematical precision, to force this triage.

4.1 The Allais paradox

Consider two decision problems involving monetary prizes, measured in millions of some currency.

Problem 1				Problem 2			
	5M	1M	0		5M	1M	0
A	–	1.00	–	C	–	0.11	0.89
B	0.10	0.89	0.01	D	0.10	–	0.90

The common pattern observed in experiments — and, as we will see, even among experts who fully understand the theory — is $A \succ B$ in the first problem and $D \succ C$ in the second. Under expected utility, these two rankings translate into

$$\begin{aligned}
 A \succ B &\iff 0.11 u(1) > 0.10 u(5) + 0.01 u(0), \\
 D \succ C &\iff 0.10 u(5) + 0.01 u(0) > 0.11 u(1),
 \end{aligned}$$

which are flatly contradictory statements about the same three numbers $u(0), u(1), u(5)$, whatever function u happens to be. This is not a question of risk aversion; a suitably curved u cannot rescue the pattern, because the contradiction holds for every u whatsoever. The trouble is independence itself, and the following decomposition shows precisely where it breaks. Let Q denote the lottery “€5M with probability 10/11, €0 with probability 1/11.” Direct calculation confirms

$$\begin{aligned}
 A &= 0.11 \cdot [1M] + 0.89 \cdot [1M], & B &= 0.11 \cdot Q + 0.89 \cdot [1M], \\
 C &= 0.11 \cdot [1M] + 0.89 \cdot [0], & D &= 0.11 \cdot Q + 0.89 \cdot [0].
 \end{aligned}$$

Each problem is a comparison between the same two lotteries, $[1M]$ for certain versus Q , mixed with the same weight 0.89 against a *common consequence* — receiving €1M for certain in Problem 1, receiving nothing in Problem 2. Independence asserts that this common consequence should be irrelevant to the ranking. Empirically, it is far from irrelevant: the presence of a sure €1M in Problem 1 makes people extremely reluctant to gamble it away, an effect large enough to flip the ranking once the common consequence changes from a certain gain to a certain loss of the alternative. This is the **certainty effect**: certainty itself, not merely the expected utility of the sure option, has independent pulling power on preferences.

Aside · an inventor of the theory rejects his own answer

At a 1952 conference in Paris, Maurice Allais posed exactly these two problems to Leonard Savage over lunch. Savage — by then the leading architect of subjective expected utility — chose A in the first problem and D in the second, falling into the very pattern that

bears Allais's name. Savage later wrote that, upon reflection, he considered his own initial choice a mistake, and revised his answer to conform to independence — which corresponds to resolving question 3 above by revising the behaviour rather than the model. Allais, unsurprisingly, drew the opposite conclusion from the same experiment: for him, it was independence that stood refuted. Allais received the Nobel Memorial Prize in Economics in 1988, principally for other contributions to economic theory.

4.2 Framing effects

★ Puzzle ★ an epidemic, described two ways

Imagine an unusual disease outbreak is expected to kill 600 people. Two alternative programmes to combat the disease have been proposed. Choose between the two programmes of version (a), then between the two programmes of version (b).

Version (a). *Programme 1:* 200 people will be saved. *Programme 2:* with probability $1/3$, all 600 people will be saved; with probability $2/3$, no one will be saved.

Version (b). *Programme 3:* 400 people will die. *Programme 4:* with probability $1/3$, nobody will die; with probability $2/3$, all 600 will die.

Programmes 1 and 3 describe the exact same outcome, and so do Programmes 2 and 4 — only the wording has changed, from a “lives saved” frame to a “lives lost” frame. Yet in the original experiment by Amos Tversky and Daniel Kahneman (1981), which used the wording above nearly verbatim, the two versions were given to two *different* groups of respondents: 72% of the first group chose Programme 1 under the “saved” framing, while 78% of the second chose Programme 4 under the “die” framing — a reversal of the modal preference driven by wording alone. A reader who has just answered both versions side by side was far better placed to notice the equivalence, which is precisely why the original design kept the versions apart. Outcomes described as **gains** tend to induce caution (a preference for the sure thing); the identical outcomes described as **losses** tend to induce risk-seeking (a preference for the gamble). Such framing effects are not a laboratory curiosity: they matter directly for how medical risks are communicated to patients, how public policies are presented to voters, and how the terms of consumer contracts are worded.

A satisfactory theory of choice must therefore be explicit about what counts as a *consequence* in the first place. If a change in description genuinely changes the consequence as experienced — perhaps a “95% survival rate” feels meaningfully different to a patient than a “5% mortality rate,” even though the numbers coincide — then the two framings are not describing psychologically identical alternatives, and no axiom is being violated. If, on the other hand, descriptions change choices while every consequence that plausibly matters to the decision maker remains the same, then stable, consequence-based preferences — the entire foundation on which Sections 1–3 were built — are being violated in a much more troubling way.

4.3 Small-stakes risk aversion and Rabin's calibration theorem

Aside · a bet refused, and a hundred bets accepted

Paul Samuelson once offered a colleague a coin-flip bet: win \$200 or lose \$100. The colleague declined the single bet, but remarked that he would happily accept a sequence of a hundred such bets. Samuelson (1963) showed this combination is inconsistent for an expected-utility maximizer: anyone who rejects the single bet at every wealth level she might hold along the way must also reject the sequence of a hundred. The sequence of n bets is obtained from the sequence of $n - 1$ by adding, at each wealth level the first $n - 1$ bets can lead to, one more bet that she rejects at that wealth level; under expected utility this makes it worse than the sequence of $n - 1$, and by induction worse than not betting at all.

Matthew Rabin (2000) sharpened this observation into a full calibration result with an unsettling conclusion. Suppose an expected-utility maximizer, at *every* level of initial wealth w , turns down a fifty-fifty bet to lose \$100 or gain \$110. We show this modest, everyday act of caution has extreme consequences for the shape of u .

Theorem 4.1: Rabin's calibration theorem (illustrative case)

Suppose an expected-utility maximizer, whose utility u over final wealth is defined on $\mathbb{R}$, differentiable, concave, and strictly increasing, rejects the lottery “lose \$100 or gain \$110, each with probability $1/2$ ” at every initial wealth level $w \geq w_0$. Then, at every wealth level $W \geq w_0 + 1,370$, she also rejects the lottery “lose \$1,680 or gain G , each with probability $1/2$ ”, *however large* G is.

Proof. By Appendix A.3, u' is nonincreasing. Moreover $u' > 0$ everywhere: if $u'(y) = 0$ for some y , then $u' \leq 0$ on $[y, \infty)$, and u could not increase there.

Marginal utility decays geometrically. Rejection at w means $\frac{1}{2}u(w - 100) + \frac{1}{2}u(w + 110) \leq u(w)$, that is,

$$u(w + 110) - u(w) \leq u(w) - u(w - 100).$$

By the mean value theorem and the monotonicity of u' , the left side is at least $110 u'(w + 110)$ and the right side at most $100 u'(w - 100)$. Hence, with $\rho = 100/110 = 10/11$ and $y = w - 100$,

$$u'(y + 210) \leq \rho u'(y) \quad \text{for every } y \geq w_0 - 100 :$$

every additional \$210 of wealth cuts marginal utility by a factor of at least $10/11$.

The gain side is bounded. Fix $W \geq w_0 + 1,370$ and $G > 0$, and pick an integer K with $210K \geq G$. On each interval $[W + 210k, W + 210(k + 1)]$ the mean value theorem bounds the increase of u by $210 u'(W + 210k) \leq 210 \rho^k u'(W)$. Since u is increasing,

$$u(W + G) - u(W) \leq \sum_{k=0}^{K-1} 210 \rho^k u'(W) < \frac{210}{1 - \rho} u'(W) = 2,310 u'(W).$$

The loss side is not. Splitting $[W - 1,680, W]$ into eight intervals of length 210 and bounding u' on each from below by its value at the right end,

$$u(W) - u(W - 1,680) \geq \sum_{j=0}^7 210 u'(W - 210j) \geq 210 u'(W) \sum_{j=0}^7 \rho^{-j} > 2,401 u'(W),$$

where $u'(W - 210j) \geq \rho^{-j}u'(W)$ follows from j applications of the decay inequality, all at points $y \geq W - 1,470 \geq w_0 - 100$, and $210 \sum_{j=0}^7 (11/10)^j \approx 2,401.5$.

Conclusion. The lottery is worth $\frac{1}{2}[u(W + G) - u(W)] - \frac{1}{2}[u(W) - u(W - 1,680)] < \frac{1}{2}(2,310 - 2,401)u'(W) < 0$ relative to keeping W , so she rejects it. ■

The bookkeeping above is deliberately crude; Rabin (2000) tracks the decay of marginal utility more finely and obtains considerably more striking figures. Notice what the argument uses: nothing beyond concavity and the rejection of one small bet over a range of wealth levels — in particular, no particular functional form for u .

Refusing a small, favourable bet is entirely ordinary and, most people would agree, entirely sensible — caution about a modest loss is not a character flaw. Refusing a fifty-fifty chance of losing \$1,680 against an unlimited potential gain, by contrast, is behaviour that almost nobody displays and few would defend. Rabin's theorem shows that, for an expected-utility maximizer whose aversion to the small bet is explained by the curvature of a utility function *defined over final wealth*, the first attitude, held at all wealth levels above some threshold, forces the second. Since ordinary people display the first without the second, the theorem is usually read as an argument that expected utility over final wealth is the wrong model of small-stakes risk aversion — perhaps what is curved is not utility over total wealth but something more like utility over *gains and losses* relative to a reference point, an idea developed at length in prospect theory (Kahneman and Tversky 1979). The *Deal or No Deal* evidence of Section 2.4 points the same way: under expected utility over final wealth, only the wealth levels still attainable should matter, not the run of gains and losses by which the contestant arrived at the current offer — yet that run is precisely what Post et al. (2008) find to drive the willingness to gamble. Rabin and Richard Thaler (2001) drew the conclusion bluntly in their “Anomalies” column on risk aversion, closing with a nod to Monty Python's dead-parrot sketch: expected utility, they wrote, is an “ex-hypothesis.”

4.4 Two boxes, one predictor: Newcomb's problem

The paradoxes above each pit expected-utility theory against something else — against itself in Allais, against a single-prior model of belief in Ellsberg, against the very definition of a consequence in the framing example, against a specific functional form in Rabin. The puzzle in this section is different in kind: it pits two principles of rational choice that ordinarily agree with each other squarely against one another, and asks which one deserves our trust when, for once, they conflict.

★ Puzzle ★ open the boxes

A predictor, whom we will call Omega, is known to be correct in her predictions almost always. Omega has already placed money into two boxes and departed the scene. Box *A* is transparent and is seen to contain €1,000. Box *B* is opaque and contains either €1,000,000 or €0, the amount having been decided *before* you make any choice, according to Omega's prediction of what you will do:

- if Omega predicted you would take **only Box B**, she placed €1,000,000 inside it;
- if Omega predicted you would take **both boxes**, she left Box *B* empty.

You may take Box *B* alone, or take both boxes. What should you do?

Definition 4.2: Two competing principles of choice

Dominance. By the time you choose, the content of Box *B* is already fixed and cannot be changed by your decision — nothing about opening a box today reaches backward in time to alter what Omega already did. Whatever Box *B* contains, taking both boxes yields exactly €1,000 more than taking Box *B* alone would, for either possible content. Dominance recommends taking **both boxes**.

Expected value, conditional on the act chosen. Among people who take only Box *B*, almost all — by hypothesis about Omega’s reliability — walk away with €1,000,000; among those who take both boxes, almost all walk away with only €1,000. Judged by the expected payoff conditional on which act is chosen, taking **Box *B* alone** is overwhelmingly better.

These two principles ordinarily point the same way, and it takes a carefully engineered scenario to pull them apart. What makes Newcomb’s problem work is that your action is statistically correlated with a state of the world — Omega’s earlier prediction — that your action cannot causally influence, because the prediction was made and the boxes were filled before you ever chose. The underlying technical divide is between **causal decision theory**, which evaluates acts strictly by their causal consequences and recommends taking both boxes, and **evidential decision theory**, which evaluates acts by what choosing them is evidence for and recommends taking Box *B* alone. Unlike the earlier paradoxes in this section, this one has never converged on anything resembling a professional consensus; each camp tends to regard the other’s preferred answer as close to self-evidently mistaken.

Aside · a philosopher’s audience, evenly split

The physicist William Newcomb devised this puzzle in 1960; the philosopher Robert Nozick brought it to wide attention in a influential 1969 paper, “Newcomb’s Problem and Two Principles of Choice.” Nozick reports that whenever he presents the problem to an audience, opinion splits close to evenly between the two answers, and members of each camp are typically genuinely baffled that the other side fails to see what strikes them as the obvious resolution. Remarkably few philosophical puzzles are sharp enough to produce such a stable, reproducible split of opinion — which is part of why the problem continues to be discussed more than sixty years after it was first posed.

In summary

The Allais paradox attacks independence directly; Ellsberg’s urn attacks the adequacy of a single additive subjective prior; framing effects attack the very notion of a well-defined consequence; Rabin’s theorem attacks the curvature-based explanation of risk aversion at small stakes; and Newcomb’s problem attacks the presumption that “maximize expected utility” picks out one uniquely correct act, once actions can be statistically correlated with — without causally influencing — the states that determine their payoff. Each of these attacks is genuinely informative precisely *because* the underlying axioms are stated with enough precision to be falsified. A theory too vague to be pinned down this way could never be refuted by any observation — and, for exactly the same reason, could never teach us anything either.

5 A sequential decision: when to stop looking

Every example so far has involved a single, one-shot choice. Many important decisions instead unfold over time: information arrives gradually, earlier options disappear once declined, and the decision maker must judge not only *what* to choose but *when* to stop gathering information and commit. Problems of this kind are called **optimal stopping problems**. This section works through one completely solvable example in detail, less for its practical advice — which turns out to be far narrower than its popular reputation suggests — than for what it teaches about how heavily an “optimal” recommendation can depend on the exact assumptions feeding into it.

5.1 The model

Picture a decision maker who will meet a sequence of n candidates, one at a time, and must decide about each one, immediately and irrevocably, whether to select them or move on. This is traditionally called the *secretary problem*, after an early application to hiring, though the same mathematics applies to searching for an apartment, an academic job offer, or, as we will explore below, a life partner. The assumptions are:

- the n candidates arrive in a uniformly random order;
- after meeting a candidate, the decision maker can rank that candidate relative to all previously met candidates, but has no absolute scale of quality — only *ordinal* information is available;
- a candidate who is passed over is gone for good — there is **no recall**;
- the sole objective is to end up with *the very best* of the n candidates; ending up with the second-best, or any other candidate, counts as complete failure.

Definition 5.1: Cutoff rule

The **cutoff rule** with parameter r rejects the first r candidates unconditionally — call this the *exploration phase* — and thereafter accepts the first subsequent candidate who ranks better than every candidate seen so far.

Theorem 5.2: Success probability of the cutoff rule

For any $1 \leq r < n$, the cutoff rule with parameter r selects the single best of the n candidates with probability

$$P_n(r) = \frac{r}{n} \sum_{k=r+1}^n \frac{1}{k-1}.$$

Proof. Condition on the position k at which the overall best candidate happens to arrive; by the uniformly random arrival order, each position $k \in \{1, \dots, n\}$ is equally likely, with probability $1/n$.

If $k \leq r$, the best candidate arrives during the unconditional exploration phase and is automatically rejected; the rule cannot possibly succeed in this case.

If $k > r$, the rule selects the best candidate exactly when it has not stopped before position k , that is, when none of the candidates in positions $r+1, \dots, k-1$ beats everyone who arrived

before them. Consider the best among the first $k - 1$ arrivals, at some position j . If $j > r$, that candidate beats everyone before them, so the rule stops at or before position $j < k$ (possibly even earlier, at a lesser record): failure. If $j \leq r$, every candidate in positions $r + 1, \dots, k - 1$ is beaten by the candidate at position j , so none of them is accepted; the rule is still running at position k , where it accepts the overall best: success.

So: the rule succeeds when the best candidate is at position k *and* the best among the first $k - 1$ candidates falls within the first r positions. By symmetry (each of the $k - 1$ earlier arrivals is equally likely to be the best among them), this second event has probability $r/(k - 1)$. Since the position of the overall best (equal to $1/n$ for landing exactly at k) and the relative rank of the first $k - 1$ arrivals are independent under a uniformly random arrival order, the probability of both holding simultaneously is $\frac{1}{n} \cdot \frac{r}{k-1}$. Summing over all valid positions $k = r + 1, \dots, n$ for the overall best candidate gives

$$P_n(r) = \sum_{k=r+1}^n \frac{1}{n} \cdot \frac{r}{k-1} = \frac{r}{n} \sum_{k=r+1}^n \frac{1}{k-1},$$

as claimed. ■

5.2 The 37% rule

To find the best cutoff r , treat n as large and set $x = r/n \in (0, 1)$. Approximating the sum in Theorem 5.2 by an integral (Appendix A.5 makes this precise),

$$P_n(r) \approx x \int_x^1 \frac{dt}{t} = x \log \frac{1}{x} =: P(x).$$

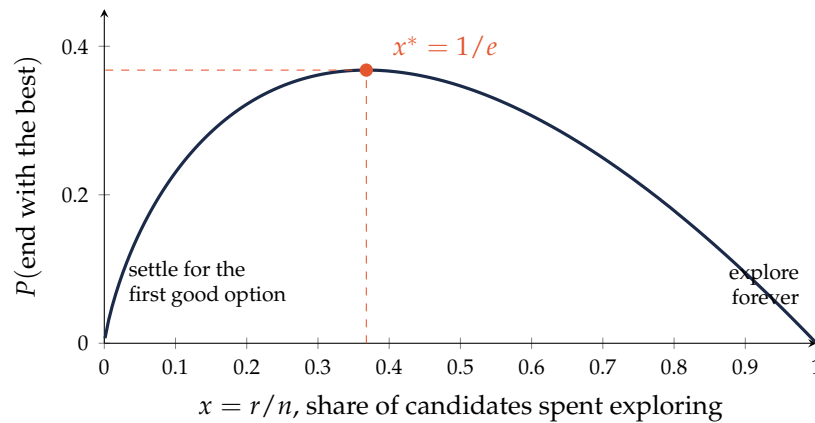
Differentiating,

$$P'(x) = \log \frac{1}{x} - 1,$$

using the product rule and $\frac{d}{dx} \log(1/x) = -1/x$, so $\frac{d}{dx} [x \log(1/x)] = \log(1/x) + x \cdot (-1/x) = \log(1/x) - 1$. Setting $P'(x) = 0$ gives $\log(1/x) = 1$, i.e. $x = 1/e \approx 0.368$; since $P''(x) = -1/x < 0$, this critical point is a maximum (see Appendix A.6), and indeed the global one, since P' is positive on $(0, 1/e)$ and negative on $(1/e, 1)$. The limiting success probability is

$$P(1/e) = \frac{1}{e} \log(e) = \frac{1}{e} \approx 36.8\%,$$

in the limit as n grows. For finite n the best achievable probability is somewhat higher — 43.3% for $n = 5$, 39.9% for $n = 10$, 37.1% for $n = 100$ — and it approaches $1/e$ from above; the remarkable fact is that it does not fall to zero, however large the field. Reject roughly the first 37% of all candidates unconditionally, purely to calibrate a sense of the field, and then commit to the very next candidate who outshines everyone seen so far. This is the “37% rule” of popular accounts such as Christian and Griffiths (2016).



★ Puzzle ★ putting numbers on it

If you anticipate meeting $n = 10$ serious candidates — job offers, apartments, or life partners, the mathematics does not care which — the exact optimum (from Theorem 5.2, computed directly rather than via the large- n approximation) is $r = 3$, achieving success probability $P_{10}(3) \approx 39.9\%$. This is not simply the mechanical rounding of $10/e \approx 3.68$ down to the nearest integer: computing the neighbouring values directly gives $P_{10}(2) \approx 36.6\%$ and $P_{10}(4) \approx 39.8\%$, so $r = 4$ performs almost as well as $r = 3$, while $r = 2$ is noticeably worse — the optimum is exact, but the function $P_{10}(\cdot)$ is fairly flat near its peak. Reject the first three candidates on principle, then commit to the first one after that who beats all three.

If the natural unit is time rather than a headcount — candidates arriving uniformly over some window, such as searching for a partner between ages 18 and 40 — the same reasoning applies directly to elapsed time. Stop exploring and start committing at age $18 + 0.368 \times (40 - 18) \approx 26$.

The rule's dark side. Even played optimally, the rule fails about 63% of the time, and it fails in two distinct ways. If the overall best candidate arrives during the exploration phase — probability $r/n \approx 37\%$ — then nobody after the exploration phase beats everyone seen so far, and the rule never accepts anyone; under the usual convention of taking the last candidate when nobody else is left, it ends with whoever happens to come last. If the best arrives later, the rule may still stop too early, at a candidate who beats everyone seen so far without being the best overall; for large n this happens with probability about $1 - 2/e \approx 26\%$. The total failure probability is $1 - 1/e \approx 63\%$: the strategy misses the single best candidate almost twice as often as it finds it.

Aside · an astronomer's second marriage

After the death of his first wife in 1611, the astronomer Johannes Kepler methodically evaluated eleven candidates for a second marriage over roughly two years, corresponding with go-betweens and weighing each prospect's merits in detail. He hesitated over one candidate, reconsidered an earlier one he had initially passed over, and eventually married the fifth candidate on his list. Thomas Ferguson (1989), in a historical survey of the secretary problem, points out that Kepler's willingness to reconsider an earlier candidate directly violates the no-recall assumption — a freedom the mathematical model denies its decision maker entirely.

5.3 Why the 37% rule is not really relationship advice

Caveat · worth reading twice

The elegance of the 37% rule should not obscure how narrow its assumptions are. Change any one of them, and the “optimal” behaviour changes with it.

- **Only the single best candidate counts as success.** If the objective is instead to minimize the *expected rank* of whichever candidate is eventually chosen — treating second-best as a fine outcome, not a catastrophe — the optimal policy no longer uses one fixed cutoff at all: it uses a sequence of increasingly demanding thresholds as the search progresses, and it achieves an expected rank of only about 3.87 even as $n \rightarrow \infty$ (Chow, Moriguti, Robbins, and Samuels, 1964). Aiming for “good enough” turns out to be a dramatically easier target, in a precise mathematical sense, than aiming for “the very best.”
- **Only ordinal information is available.** If instead the decision maker observes each candidate’s quality on an absolute scale, the qualities being independent draws from a *known* distribution — a numerical test score rather than a mere local rank — she can succeed in finding the true best with probability closer to 58% (Gilbert and Mosteller, 1966), far better than $1/e$.
- **The total number of candidates, n , is known in advance.** Nobody embarking on a genuine search for a life partner, or even a job, typically knows this number with any precision.
- **The choice is entirely one-sided.** The model treats the searcher as the only party making a decision, but in genuine courtship, hiring, or house-hunting, the other side is simultaneously solving its own stopping problem and may decline to be selected. A fully two-sided search of this kind is a matching problem rather than a stopping problem, with a mathematical theory of its own — the theory of stable matching, developed by David Gale and Lloyd Shapley, which sits well beyond the scope of this chapter.

The value of working through this example carefully lies not in the 37% figure itself, but in seeing exactly how an apparently practical, quotable recommendation is inseparable from a long list of specific informational and structural assumptions. Change the objective from “the very best” to “good enough,” or change the information from purely ordinal to fully cardinal, or make the search two-sided instead of one-sided, and the mathematically optimal behaviour changes right along with it. This is, in miniature, the central lesson of decision theory as a whole: rigor buys precision about what follows from stated assumptions, never a shortcut past the need to state them.

6 What this chapter establishes

In summary

- Individual choice is fully specified by a feasible set, a preference relation, and a choice rule — and confusing these three objects is one of the most common sources of avoidable disagreement.
- Completeness and transitivity make coherent ranking possible; a utility function represents that ranking numerically — in an ordinal sense only, unless the model supplies further information, such as comparisons of preference intensities.
- Under risk, adding continuity and independence yields the expected-utility representation, with a Bernoulli utility function that is cardinal — unique up to positive affine transformations, not merely up to arbitrary relabeling.
- Concavity of the Bernoulli utility function is exactly equivalent to risk aversion; it generates the entire apparatus of certainty equivalents, risk premia, and the value of insurance.
- Under genuine uncertainty, a single subjective probability distribution may be too narrow an object to represent every coherent attitude toward ambiguity. Maxmin expected utility (a set of priors) and Choquet expected utility (a non-additive capacity) are two responses, and they coincide when the capacity is convex.
- The Allais paradox, Ellsberg's urn, framing effects, Rabin's calibration theorem, and Newcomb's problem are each informative precisely *because* the axioms they challenge are stated sharply enough to be tested and, in each case, found wanting somewhere.
- Sequential decision problems add irreversibility and the value of information over time; any claim that a particular stopping rule is “optimal” is only ever optimal relative to a specific, and often quite narrow, set of assumptions.

The concepts built in this chapter will be reused throughout the rest of the book without being reintroduced from scratch. The natural next step is to replace a single decision maker with many, each bringing their own preferences to a shared decision. The central difficulty then is no longer uncertainty about consequences, but disagreement about how several individual rankings ought to determine a single collective outcome — and Kevin's money-pumping cycle from the very beginning of this chapter will make a return appearance, this time generated collectively by a group of voters every one of whom, individually, has entirely transitive preferences.

6.1 Key definitions

Feasible set

The alternatives available in a particular decision problem.

Weak preference

A binary relation $\succsim$ read as “at least as good as.”

Weak order

A complete and transitive binary relation.

Utility representation

A function u satisfying $x \succsim y \iff u(x) \geq u(y)$.

Choice correspondence

A rule assigning a nonempty set of selected alternatives to every nonempty feasible set.

Condition α

An alternative chosen from a menu remains chosen from any smaller menu that contains it.

Pareto efficiency

No feasible alternative is weakly better on every criterion and strictly better on at least one.

Lottery

A probability distribution over consequences.

Expected utility

The probability-weighted average of Bernoulli utilities.

Risk aversion

Weakly preferring the expected value of any lottery, received for certain, to the lottery itself; equivalent, under expected utility, to concavity of the Bernoulli utility function.

Certainty equivalent

The certain wealth level yielding the same utility as a given lottery.

Risk premium

Expected wealth minus the certainty equivalent.

Act

A mapping from states of the world to consequences.

Ambiguity aversion

A preference for well-specified risks over poorly specified ones, other things held fixed.

Capacity

A monotone set function on events, equal to 0 on the empty event and 1 on the sure event, that need not be additive.

Choquet integral

The evaluation of an act with respect to a capacity, weighting each increment of utility by the capacity of the event on which it is obtained.

Optimal stopping problem

A sequential decision problem in which one must choose when to stop gathering information and commit to an irreversible action.

6.2 Exercises

- E1.** Show that if u represents a preference relation $\succsim$ in expected-utility form, then u^3 generally does not. (Hint: find two lotteries ranked one way by $\mathbb{E}[u]$ and the other way by $\mathbb{E}[u^3]$; a two-outcome example suffices.)
- E2.** Compute the certainty equivalent of the St. Petersburg game (Section 2.2) directly for $u(x) = \sqrt{x}$, and compare it to the certainty equivalent computed in the text for $u(x) = \log x$.
- E3.** Prove directly from the definitions that the Arrow–Pratt coefficient $A(w)$ is invariant under the transformation $u \mapsto au + b$ for $a > 0$, and compute both $A(w)$ and $R(w)$ for $u(w) = \log w$.

- E4.** In the evening example of Section 1.5, add a fourth option D , a concert, scoring $(8, 5)$ on (enjoyment, rest). Is D Pareto efficient? Is there any weight w for which D maximizes the weighted score S_w ? Interpret the answer geometrically.
- E5.** For $n = 5$ candidates, compute $P_5(r)$ from Theorem 5.2 for every value of $r \in \{1, 2, 3, 4\}$, and confirm directly that $r = 2$ is optimal.
- E6.** Explain, in your own words and without symbols, why the modal choice pattern in the Allais paradox violates independence but does not violate transitivity. (Hint: transitivity is a condition on a single preference relation over a fixed set of options; what exactly changes between Problem 1 and Problem 2?)
- E7.** In the Newcomb problem, suppose Omega's prediction is correct with probability q , whatever you choose, rather than "almost always." Compute the expected payoff from each strategy as a function of q , find the accuracy threshold above which taking Box B alone yields a strictly higher expected payoff than taking both boxes, and evaluate both payoffs at $q = 0.9$. Why would a causal decision theorist be unmoved by this calculation?
- E8.** Consider the Ellsberg capacity v of Section 3.3. (a) Check that v is convex. (b) Let h be the act yielding utility 0 if the ball is red, 2 if black, and 1 if yellow. Compute $\int h dv$ and compare it with $\min_{P \in \mathcal{P}} \mathbb{E}_P[h]$. (c) Let $\bar{v}(E) = \max_{P \in \mathcal{P}} P(E)$. Is $\bar{v}$ convex? Rank the four Ellsberg bets by their Choquet values under $\bar{v}$, and interpret.

Selected solutions. E2: $\mathbb{E}[\sqrt{X}] = \sum_k 2^{-k} 2^{k/2} = \sum_k (1/\sqrt{2})^k = (1/\sqrt{2})/(1 - 1/\sqrt{2}) = 1/(\sqrt{2} - 1)$, so the certainty equivalent is $c = (1/(\sqrt{2} - 1))^2 = (\sqrt{2} + 1)^2 = 3 + 2\sqrt{2} \approx 5.83$ euros — higher than the €4 found for log utility, because $\sqrt{\cdot}$ is less concave (less risk averse) than $\log(\cdot)$ at these wealth levels. E3: $A(w) = 1/w$, $R(w) = 1$ for every w . E4: D is Pareto efficient (A beats it on enjoyment only, B and C on rest only), yet $S_w(D) = 5 + 3w$ never maximizes the score: for $w < 2/3$ it is below $S_w(B) = 7$, and for $w \geq 2/3$ below $S_w(A) = 4 + 5w$, the gap being $2w - 1 > 0$. Geometrically, D lies strictly below the segment joining A and B in the (enjoyment, rest) plane — the midpoint of that segment is $(8, 5.5)$ — and weighted scores can only select efficient options on the boundary of the convex hull of the feasible score vectors. E5: $P_5(1) = 0.417$, $P_5(2) = 0.433$, $P_5(3) = 0.350$, $P_5(4) = 0.200$, confirming $r = 2$ is optimal for $n = 5$. E7: with predictor accuracy q , taking Box B alone yields expected payoff $q \cdot 1,000,000 + (1 - q) \cdot 0$, while taking both boxes yields $q \cdot 1,000 + (1 - q) \cdot 1,001,000$; setting these equal and solving gives a threshold accuracy of $q^* = 1,001,000/2,000,000 = 0.5005$ — just over 50% — above which Box B alone has the higher expected payoff; at $q = 0.9$ the two payoffs are €900,000 and €101,000. The evidential recommendation thus requires only a predictor slightly better than a coin flip, not near-infallibility. The calculation, however, conditions on the act chosen: a causal decision theorist holds the content of Box B fixed, whatever its probability, notes that taking both boxes adds €1,000 in every case, and is unmoved by any value of q . E8: (a) when $E \subseteq F$ or $F \subseteq E$ the inequality is an equality; the remaining pairs are checked directly, e.g. $v(B \cup Y) + v(\emptyset) = \frac{2}{3} \geq v(B) + v(Y) = 0$ and $v(S) + v(B) = 1 \geq v(R \cup B) + v(B \cup Y) = 1$. (b) Ordering black, yellow, red, $\int h dv = (2 - 1)v(B) + (1 - 0)v(B \cup Y) = \frac{2}{3}$, while $\mathbb{E}_P[h] = 2b + (\frac{2}{3} - b) = \frac{2}{3} + b$ is minimized at $b = 0$, with the same value, as Theorem 3.5 predicts. (c) $\bar{v}(B) = \bar{v}(Y) = \bar{v}(B \cup Y) = \frac{2}{3}$, so $\bar{v}(B \cup Y) + \bar{v}(\emptyset) < \bar{v}(B) + \bar{v}(Y)$: $\bar{v}$ is not convex. The bets are valued $\bar{v}(R) = \frac{1}{3} < \frac{2}{3} = \bar{v}(B)$ and $\bar{v}(R \cup Y) = 1 > \frac{2}{3} = \bar{v}(B \cup Y)$: the opposite of the Ellsberg pattern, as for a decision maker who always expects the most favourable composition of the urn — an ambiguity lover.

6.3 Further reading

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A Mathematical prerequisites

This appendix collects, with proofs, the pure mathematics used in the chapter beyond elementary algebra: order relations and their properties, countability, concavity and Jensen's inequality, and the calculus facts (Taylor's theorem, the comparison of sums with integrals, and second-derivative tests) used in Sections 2–5. Only the mean value theorem and the elementary properties of the integral are taken for granted. Reading it before the chapter is one way through the material; consulting it only when a proof refers to it is another.

A.1 Order relations

Given a set X , recall that a binary relation $\succsim$ on X is simply a specification, for every ordered pair $(x, y) \in X \times X$, of whether $x \succsim y$ holds or not. Three properties of such relations recur constantly.

Definition A.1: Reflexivity, completeness, transitivity

A binary relation $\succsim$ on X is:

- **reflexive** if $x \succsim x$ for every $x \in X$;
- **complete** if for every $x, y \in X$, $x \succsim y$ or $y \succsim x$ (or both);
- **transitive** if for every $x, y, z \in X$, $x \succsim y$ and $y \succsim z$ together imply $x \succsim z$.

Completeness on its own already implies reflexivity: apply the definition to the pair (x, x) to get $x \succsim x$ or $x \succsim x$, i.e. $x \succsim x$. A relation that is complete and transitive is called a **weak order**; this is the technical name, used throughout the chapter, for what an ordinary reader would simply call a coherent ranking with ties allowed.

From a weak order $\succsim$ we build the **strict part** $\succ$ and the **indifference part** $\sim$ by

$$x \succ y \iff x \succsim y \text{ and not } y \succsim x, \quad x \sim y \iff x \succsim y \text{ and } y \succsim x.$$

It is a short exercise, used implicitly throughout, that $\sim$ is itself reflexive, transitive, and symmetric (an *equivalence relation*), and that $\succ$ is irreflexive and transitive (a **strict partial order**). It is *not* total — two indifferent alternatives are ranked by $\succ$ in neither direction — but it has the stronger property of **negative transitivity**: if neither $x \succ y$ nor $y \succ z$, then not $x \succ z$. A relation with these properties is called a **strict weak order**.

A.2 Countability and why the lexicographic order has no utility function

A set S is **countable** if its elements can be listed in a sequence $s_1, s_2, s_3, \dots$ (possibly finite, possibly infinite) — equivalently, if there is a function from S onto a subset of the natural numbers. The rational numbers $\mathbb{Q}$ are countable (a standard diagonal enumeration lists all fractions p/q), while the real numbers $\mathbb{R}$ are famously **not** countable, by Cantor's diagonal argument. A consequence used in the chapter's discussion of the lexicographic order: any collection of pairwise disjoint, nondegenerate (more than a single point) open intervals of real numbers must be countable, because each such interval contains at least one rational number, distinct intervals contain distinct rationals (disjointness), and a set that injects into the countable set $\mathbb{Q}$ is itself countable.

A.3 Concavity and Jensen's inequality

Definition A.2: Concave and convex functions

A function $u : \mathbb{R} \rightarrow \mathbb{R}$ (or defined on an interval of $\mathbb{R}$) is **concave** if, for every x, y in its domain and every $\lambda \in [0, 1]$,

$$u(\lambda x + (1 - \lambda)y) \geq \lambda u(x) + (1 - \lambda)u(y).$$

It is **strictly concave** if the inequality is strict whenever $x \neq y$ and $\lambda \in (0, 1)$; **convex** if the reverse inequality holds; and **affine** if equality holds always (an affine function has the form $u(x) = ax + b$).

Geometrically, concavity says the graph of u lies above every chord connecting two of its points; this is exactly the “bends downward, like the graph of $\sqrt{x}$ ” picture used informally throughout the chapter.

Theorem A.3: Slopes of a concave function

Let u be concave on an interval I , and let $x < z < y$ be points of I . Then

$$\frac{u(z) - u(x)}{z - x} \geq \frac{u(y) - u(x)}{y - x} \geq \frac{u(y) - u(z)}{y - z}.$$

Conversely, if the outer inequality (first expression $\geq$ last) holds for all $x < z < y$ in I , then u is concave.

Proof. Write $z = \lambda x + (1 - \lambda)y$ with $\lambda = (y - z)/(y - x) \in (0, 1)$, so that $z - x = (1 - \lambda)(y - x)$ and $y - z = \lambda(y - x)$. Concavity gives $u(z) \geq \lambda u(x) + (1 - \lambda)u(y)$, hence $u(z) - u(x) \geq (1 - \lambda)(u(y) - u(x))$ and $u(y) - u(z) \leq \lambda(u(y) - u(x))$; dividing by $z - x$ and $y - z$ respectively gives the two inequalities. Conversely, the outer inequality reads $\lambda(u(z) - u(x)) \geq (1 - \lambda)(u(y) - u(z))$, i.e. $u(z) \geq \lambda u(x) + (1 - \lambda)u(y)$; as $x < y$ and $\lambda \in (0, 1)$ are arbitrary (the cases $\lambda \in \{0, 1\}$ being trivial), u is concave. ■

Theorem A.4: Derivative tests for concavity

Let u be differentiable on an open interval I .

- (i) u is concave on I if and only if u' is nonincreasing on I .
- (ii) If u is twice differentiable on I , then u is concave on I if and only if $u''(x) \leq 0$ for every $x \in I$; if $u'' < 0$ throughout I , then u is strictly concave.

Proof. (i) Let u be concave and $x < y$ in I . Letting $z \downarrow x$ in the first inequality of Theorem A.3 gives $u'(x) \geq \frac{u(y) - u(x)}{y - x}$, and letting $z \uparrow y$ in the second gives $\frac{u(y) - u(x)}{y - x} \geq u'(y)$; hence $u'(x) \geq u'(y)$. Conversely, let u' be nonincreasing and $x < z < y$. The mean value theorem gives $\xi_1 \in (x, z)$ and $\xi_2 \in (z, y)$ with

$$\frac{u(z) - u(x)}{z - x} = u'(\xi_1) \geq u'(\xi_2) = \frac{u(y) - u(z)}{y - z},$$

which is the outer inequality; u is concave by Theorem A.3.

(ii) If $u'' \leq 0$ on I , the mean value theorem applied to u' gives $u'(y) - u'(x) = u''(\xi)(y - x) \leq 0$ for $x < y$, so u' is nonincreasing. Conversely, if u' is nonincreasing, its difference quotients

are nonpositive, and so is their limit $u''(x)$. Part (i) concludes. If $u'' < 0$ throughout, the same argument shows that u' is strictly decreasing, the displayed inequality in (i) becomes strict, and so does the resulting concavity inequality. ■

For $u(x) = \sqrt{x}$ on $(0, \infty)$: $u'(x) = \frac{1}{2}x^{-1/2}$ and $u''(x) = -\frac{1}{4}x^{-3/2} < 0$ for every $x > 0$, so u is strictly concave on its domain, as used in Section 2.4.

Theorem A.5: Jensen's inequality (finite version)

If u is concave and $p_1, \dots, p_m \geq 0$ satisfy $\sum_i p_i = 1$, then for any $x_1, \dots, x_m$ in the domain of u ,

$$u\left(\sum_{i=1}^m p_i x_i\right) \geq \sum_{i=1}^m p_i u(x_i).$$

Proof. By induction on m . The case $m = 1$ is trivial ($p_1 = 1$), and the case $m = 2$ is exactly the definition of concavity above. Suppose the statement holds for some $m \geq 2$; we prove it for $m + 1$ points. Given weights $p_1, \dots, p_{m+1}$ summing to one (assume $p_{m+1} < 1$, else the statement is trivial), let $q_i = p_i / (1 - p_{m+1})$ for $i = 1, \dots, m$, so that $q_1, \dots, q_m \geq 0$ sum to one. By the inductive hypothesis applied to these m points and weights,

$$u\left(\sum_{i=1}^m q_i x_i\right) \geq \sum_{i=1}^m q_i u(x_i).$$

Now apply the two-point definition of concavity to the pair of points $\bar{x} := \sum_{i=1}^m q_i x_i$ and x_{m+1} , with weights $(1 - p_{m+1})$ and p_{m+1} :

$$u\left((1 - p_{m+1})\bar{x} + p_{m+1}x_{m+1}\right) \geq (1 - p_{m+1})u(\bar{x}) + p_{m+1}u(x_{m+1}).$$

The left-hand argument simplifies: $(1 - p_{m+1})\bar{x} = (1 - p_{m+1})\sum_i q_i x_i = \sum_{i=1}^m p_i x_i$ by definition of q_i , so the left side is exactly $u(\sum_{i=1}^{m+1} p_i x_i)$. The right-hand side, using the inductive bound on $u(\bar{x})$, satisfies

$$(1 - p_{m+1})u(\bar{x}) + p_{m+1}u(x_{m+1}) \geq (1 - p_{m+1})\sum_{i=1}^m q_i u(x_i) + p_{m+1}u(x_{m+1}) = \sum_{i=1}^{m+1} p_i u(x_i).$$

Chaining the two displayed inequalities gives $u(\sum_{i=1}^{m+1} p_i x_i) \geq \sum_{i=1}^{m+1} p_i u(x_i)$, completing the induction. ■

This is exactly the inequality used to prove Theorem 2.6 (risk aversion equals concavity) in Section 2.4: a lottery with finitely many outcomes is exactly a finite weighted average of the kind Jensen's inequality describes.

A.4 Taylor's theorem

Theorem A.6: Taylor's theorem (second order, Peano remainder)

Let u be differentiable on an open interval containing w , and twice differentiable at w . Then

$$u(w + h) = u(w) + h u'(w) + \frac{1}{2} h^2 u''(w) + R(h), \quad \text{where } R(h)/h^2 \rightarrow 0 \text{ as } h \rightarrow 0,$$

which we abbreviate as $R(h) = o(h^2)$.

Proof. The function $R(h) = u(w+h) - u(w) - hu'(w) - \frac{1}{2}h^2u''(w)$ is differentiable near 0, with $R(0) = 0$ and $R'(h) = u'(w+h) - u'(w) - hu''(w)$. By definition of $u''(w)$, $R'(h)/h \rightarrow 0$ as $h \rightarrow 0$. By the mean value theorem, $R(h) = R(h) - R(0) = hR'(\theta h)$ for some $\theta \in (0,1)$ depending on h , so

$$\left| \frac{R(h)}{h^2} \right| = \theta \left| \frac{R'(\theta h)}{\theta h} \right| \leq \left| \frac{R'(\theta h)}{\theta h} \right| \rightarrow 0 \quad (h \rightarrow 0).$$

This is the expansion used in the proof of Theorem 2.8.

A.5 Sums and integrals: the approximation used in Section 5.2

Theorem A.7: Comparing a harmonic sum with an integral

For integers $1 \leq r < n$,

$$\log \frac{n}{r} \leq \sum_{j=r}^{n-1} \frac{1}{j}, \quad \text{and, if } r \geq 2, \quad \sum_{j=r}^{n-1} \frac{1}{j} \leq \log \frac{n-1}{r-1}.$$

Proof. Since $t \mapsto 1/t$ is decreasing, $\frac{1}{j} \geq \int_j^{j+1} \frac{dt}{t}$ and, for $j \geq 2$, $\frac{1}{j} \leq \int_{j-1}^j \frac{dt}{t}$. Summing over $j = r, \dots, n-1$ gives $\sum_{j=r}^{n-1} \frac{1}{j} \geq \int_r^n \frac{dt}{t} = \log \frac{n}{r}$ and, for $r \geq 2$, $\sum_{j=r}^{n-1} \frac{1}{j} \leq \int_{r-1}^{n-1} \frac{dt}{t} = \log \frac{n-1}{r-1}$. ■

The sum in Theorem 5.2 is $\sum_{k=r+1}^n \frac{1}{k-1} = \sum_{j=r}^{n-1} \frac{1}{j}$, so for $2 \leq r < n$,

$$\frac{r}{n} \log \frac{n}{r} \leq P_n(r) \leq \frac{r}{n} \log \frac{n-1}{r-1}.$$

If $n \rightarrow \infty$ and $r/n \rightarrow x \in (0,1)$, then $(n-1)/(r-1) \rightarrow 1/x$ as well, and both bounds converge to $x \log(1/x)$: this is the approximation $P_n(r) \approx x \log(1/x)$ used in the text. The same squeeze works for any monotone function — a sum of its values at consecutive integers lies between two integrals that differ by at most one term — which is the idea behind Riemann sums.

A.6 Classifying critical points with the second derivative

Theorem A.8: Second-derivative test

Let f be twice differentiable on an open interval and suppose $f'(x^*) = 0$ at some interior point x^* . If $f''(x^*) < 0$, then x^* is a strict local maximum of f ; if $f''(x^*) > 0$, it is a strict local minimum.

Proof. By Theorem A.6 and $f'(x^*) = 0$, $f(x^*+h) - f(x^*) = h^2(\frac{1}{2}f''(x^*) + R(h)/h^2)$ with $R(h)/h^2 \rightarrow 0$. If $f''(x^*) < 0$, the bracket is negative for all small $h \neq 0$, so $f(x^*+h) < f(x^*)$. The other case is symmetric. ■

This is the tool used to confirm that $x = 1/e$ maximizes $P(x) = x \log(1/x)$ in Section 5.2 (there $P''(x) = -1/x < 0$ everywhere on $(0,1)$, so the unique critical point is a local maximum;

it is the global one because $P'(x) = \log(1/x) - 1$ is positive to its left and negative to its right), and it is used implicitly whenever a claim in the text asserts that a particular first-order condition identifies an optimum rather than merely a critical point.